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Standard

Traffic control at work sites

Technical Manual

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Including amendments to Traffic control at work sites:

- **Amendment 1: TD 00003:2022, dated 16 November 2022**
- **Amendment 2: TD 00031:2022, dated 4 November 2022**



For queries regarding this document
standards@transport.nsw.gov.au
www.transport.nsw.gov.au

Technical Direction – TD 00003:2022

Issue date: 16 November 2022

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Title: Traffic control at work sites

This technical direction is issued by the Asset Management Branch (AMB), formerly known as the Asset Standards Authority (ASA), as an update to *Traffic control at work sites*, version 6.1.

The update includes amendments to the following provisions:

- providing for pedestrians
- sign posting traffic hazards before rectification works can take place
- sequence for installation and removal of signs and devices:
 - use of automated work vehicles as work site plant, including automatic cone placement trucks
 - motorways and other high-risk, multi-lane environments
- projected pavement arrows
- minor change to table caption
- traffic guidance scheme approval personnel.

1 **Amendment to *Traffic control at work sites*, v6.1**

The following sections in *Traffic control at work sites* are to be amended as follows:

Section 4.4.2 Pedestrians

Replace the Requirement column for Footpaths in Table 4–4 as follows:

Table 4–4 – Planning considerations for pedestrians

Consideration	Requirement
Footpaths	Temporary footpaths must provide a clear path of travel and must be: • Adequately and clearly signposted to support pedestrian wayfinding to /from the existing pedestrian facilities. • Of all weather standard including ramps over gutters that can have water velocities above 1 m/s. • Of equivalent material, surface condition and performance to the adjacent footpaths and not pose a trip hazard for the range of pedestrians. • At local constrictions, not less than 1 m width. Elsewhere a width of at least 2 m must be provided and any additional width to aid stopping sight distance to all road users. Additional width should be provided where this is a point of concentration or peak flows of pedestrians, such as in front of shops and schools, at bus and light-rail stops, railway stations and at pedestrian crossing points in medians. TTM signs and devices must not obstruct or reduce the width of an existing or temporary footpath to less than 1 m wide unless an alternative facility is provided for pedestrians.

Section 4.4.4 Motorcyclists

Replace the requirement for road surfaces, grooved roads and other hazards in their entirety with the following:

Table 4–6 – Considerations when planning for works impacting motorcycles

Consideration	Requirement
Road surfaces	Surface changes can be critical to motorcycle stability and so, surface changes must be signposted well in advance so that a motorcyclist is not forced to take rapid or unexpected evasive action. This may include: • Use of the GRAVEL ROAD (T3-13) sign where sections of sealed roads are temporarily reduced to gravel. • Use of other surface condition signs such as ROUGH SURFACE (T3-7), Loose stones symbolic (T3-9), LOOSE SURFACE (T3-14) or POTHOLE symbolic (T3-219n). • Use of the SLIPPERY symbolic (T3-3) sign to indicate a wet surface. • Use of TRAFFIC HAZARD (T1-10) sign when there is no other more relevant sign. • Use of appropriate warning signage prior to and after the hazard, such as the TRAFFIC HAZARD AHEAD (T1-10-1n) sign and the END TRAFFIC HAZARD (T1-10-2n) sign. • Use of appropriate signage, such as Temporary Portable Rumble Strip Ahead symbolic (T5-210n) sign or Boom Barrier AHEAD (T1-272n) sign to warn of specific traffic management changes.
Grooved roads	Milling of roads prior to resurfacing can present a hazard to motorcyclists. Where road surfaces are grooved, CYCLE HAZARD GROOVED ROAD (T2-207n) signs must be erected on all approaches. Where grooving is carried out on roads with curved alignments, it may be necessary to impose roadwork speed zones to help improve motorcycle safety.
Other hazards	• Materials, installation and maintenance of steel plates (road plates) used to cover excavations must comply with the QA Specification M209. • Water which drains from roadworks, should not run across traffic lanes to a depth greater than 5 mm.

Section 6.4 Sequence for installation and removal of signs and devices

Replace the contents of Section 6.4 in their entirety with the following:

Section 6.4.1 General

The sequence for installation and removal of signs and devices must be considered in the TMP and documented in the TGS or another site document such as a SWMS. The installation and removal of signs and devices must:

- Be undertaken in accordance with the procedures, sequence, signs and devices shown on the relevant TGS or detailed in the SWMS.
- Be carried out in a forward direction in the advance warning area, starting on the approaches to the work site and progressing towards the work area, unless shown otherwise on the TGS or documented in the SWMS.

- Not require workers to cross roads or multiple travel lanes on foot.
- Be undertaken with a work vehicle with a flashing arrow or rotating/flashing light(s) positioned between the workers and approaching traffic.
- Not commence until there has been an onsite (pre-start) review of the risk assessment and verification by the ITCP or PWZTMP qualified person that the TGS is still appropriate to implement.

The *hierarchy of controls* framework for risk management must be considered and applied for the installation and removal of TTM devices. Where appropriate, this includes the prioritised use of:

1. automated work vehicles (see Section 6.4.2)
2. pod-trucks that eliminate or reduce the need for worker/s on foot
3. worker/s on foot with a spotter.

Section 6.4.2 Use of automated work vehicles as work site plant

Automated work vehicles, such as automated cone placement trucks, may be used for the installation and removal of traffic control devices, where considered appropriate to safely undertake the works, such as at high risk sites.

Automated work vehicles can reduce or eliminate risk to workers who may be otherwise exposed to live traffic.

However, their use must be:

- risk assessed
- documented in the TMP
- included in the relevant TGS.

The use of automated work vehicles must be considered in the work site SWMS. The SWMS must include the risks introduced and proposed mitigation measures (such as, a tail vehicle to prevent traffic overtaking where the cone grabber arm encroaches into a live traffic lane) as well as the risks intended to be eliminated or substituted through the use of the plant.

Section 6.4.3 Installation

Section 6.4.3.1 General

Before work commences, the installation of signs and devices at the work site must be safely sequenced. Section 6.4.3.2 to Section 6.4.3.4. provide the general procedures for the installation of TTM signs and devices.

Section 6.4.3.2 Two-lane, two-way roads

For two-lane, two-way roads, installation should be carried out in the following order:

1. Install termination signs and devices (affected direction and side streets).
2. Install remaining signs and devices on side streets if applicable.
3. Install signs and devices in the non-working lane (unaffected direction).
4. Install signs and devices in the working lane (affected direction).

The sequence of installation should be as illustrated in Figure 6-3 in the following order:

1. Circled numeral 1: Install the termination signs when initially leaving work area, 'End Road Work/speed reinstatement' (affected direction).
2. Circled numeral 2: Use the existing road network to turn where safe to do so.
3. Circled numerals 3 to 7: Place approach signs in unaffected direction, including the PTCD (traffic controller to remain with the PTCD).
4. Circled numeral 8: Install 'End Road Work/speed reinstatement' (unaffected direction).
5. Circled numeral 9: Use the existing road network to turn where safe to do so.
6. Circled numerals 10 to 14: Place approach signs in the affected direction, including the PTCD (traffic controller to remain with PTCD).
7. Circled numerals 15 and 16: Traffic controller/s to stop traffic and taper/lane closure delineation implemented.
8. Circled numeral 17: ITCP qualified person completes drive around to confirm TGS is installed as designed.

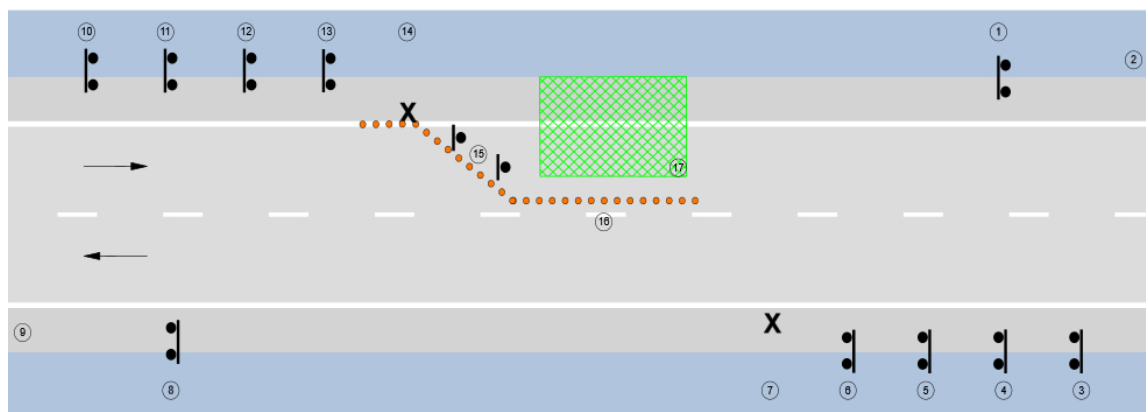


Figure 6-3 Example sign installation sequence for a two-lane, two-way road

Section 6.4.3.3 Multi-lane roads

For multi-lane roads, installation should occur in the following order:

1. Install signs and devices for the non-working lane (unaffected lane).

2. Install signs and devices for the working lane (affected lane).

The installation of signs on central medians of multi-lane divided carriageways increases the risk profile of the activity. In such cases, a site-specific TGS with either static work or a dynamic works convoy must be used.

The sequence of installation should be as illustrated in Figure 6-4 in the following order:

1. Circled numeral 1: Locate advance warning vehicle and TMA to shadow sign installation vehicle.
2. Circled numerals 2 to 5: Install advance warning signs in unaffected lane.
3. Circled numeral 6: Install 'End Roadwork'/speed reinstatement.
4. Circled numeral 7: Use the existing road network to turn where safe to do so.
5. Circled numeral 8: Locate advance warning vehicle and TMA to shadow sign installation vehicle.
6. Circled numerals 9 to 12: Install advance warning signs in obstructed (affected) lane.
7. Circled numeral 13: Install 'Flashing Arrow' and delineation devices on approach to start of taper.
8. Circled numeral 14: Position TMA in travel lane to shadow installation of taper.
9. Circled numerals 14 and 15: Install taper and delineation devices to form taper, safety buffer and past work area.
10. Circled numeral 16: Install 'End Roadwork'/speed reinstatement.
11. Circled numeral 17: Use the existing road network to turn where safe to do so.
12. Circled numeral 18: TMA positioned to shadow work area.
13. Circled numeral 19: ITCP qualified person completes drive around to confirm TGS is installed as designed.

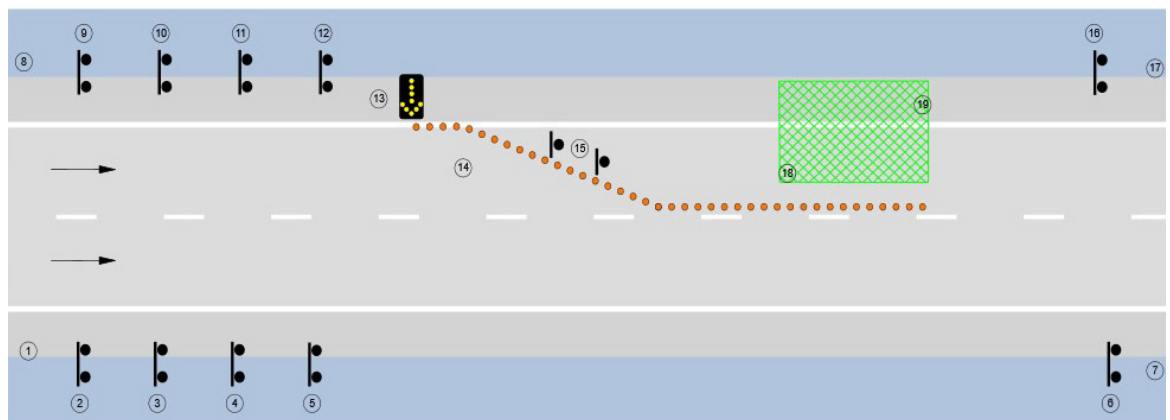


Figure 6-4 Example sign installation sequence for a multi-lane road

Where a work area is moving progressively along the road, relocation of the signs and devices located in advance of the work area should take place in accordance with the sequence described in this Section.

For long-term or recurring short-term sites, consideration should be given to the use of VMS for displaying static images that need to be repeatedly displayed during work hours and removed at the end of each shift, see Section 6.9.1.2. If static signs are used, they should be installed on posts or their location should be marked for easy placement.

Section 6.4.3.4 Motorways and other high-risk multi-lane environments

The requirements in this Section must be applied when installing signs and devices on any part of the designated roads listed in Table 6-20. These requirements should also be applied to roads not included on Table 6-20, but meet all of the following high-risk multi-lane road criteria:

- multi-lane road
- posted speed limit greater than 85 km/h
- greater than 10,000 vpd.

As part of the risk assessment process, consideration should still be given to applying these requirements on other high speed multi-lane roads even though the road may not meet the above traffic volume criteria.

The planning requirements and recommendations provided in Table 6-4 apply to the installation of signs and devices.

Table 6–4 – Planning requirements and recommendations for installation

Item	Provision/s
Risk assessment	A site-specific risk assessment must consider the risks involved in each stage of the installation process and identify the controls necessary to eliminate or minimise those risks.
TGS and documentation	A site-specific process for installation of signs and devices must be shown on the static works TGS or on a supplementary TGS by providing a numbered sequence for installation of the signs and devices. A SWMS or other generic document that applies to any location does not meet the above requirement for documenting how the static works TGS will be installed.
TGS design and approval	Both the primary TGS and (where used) the supplementary TGS for the installation of primary TGS must be approved and verified in accordance with Section 7.9.

Item	Provision/s
Co-ordination	There must be full co-ordination of any Intelligent Transport Systems (ITS) infrastructure that may assist with the TGS installation, and must be documented on the TGS. For example, this may involve the placement of advance warning messages on permanent variable message signs and/or a reduction of the speed limit where variable speed limit signs (VSLS) are provided.

The operational requirements and recommendations provided in Table 6-5 apply to the installation of signs and devices.

Table 6–5 Operational requirements and recommendations for installation

Item	Provision/s
Advance warning vehicle	At least one advanced warning vehicle must precede the TMA (acting as shadow vehicle). The advance warning vehicle must be fitted with a VMS. The VMS size must be in accordance with Section 6.5.3. During installation of the taper, or when the TMA is otherwise occupying an open trafficable lane, the advance warning vehicle should regularly communicate with heavy vehicle drivers via UHF radio and provide verbal advance warning of the TTM operation. The advanced warning vehicle must be in the road shoulder, offset from the travel lane edge line by at least 2.0 m. Offsets from the travel lane edge line of less than 2.0 m may be used, provided an additional TMA is used as the advance warning vehicle.
Shadow vehicle	A TMA must be used to shadow the traffic control vehicle and workers on foot. A TMA must not be located in the travel lane, unless: - the TMA is protecting TTM personnel who are in the process of placing signs and devices to form the taper - the TMA is located inside the lane closure (behind the taper) - the installation activity is carried out using a dynamic (mobile) works convoy in accordance with Section 7.8.4.4.
Number of vehicles	Additional, suitably sized vehicles must be in place if gaps in traffic are insufficient for the TMA to appropriately shadow the traffic control vehicle and workers on foot continually.
Duplication of signs	As part of the risk assessment process, consideration must be given to whether it is appropriate to duplicate key signs (that is, lane status, speed limit, road work ahead and so on) on the median or on the shoulder.

The installation of signs and devices on central medians on motorways and other high risk multi-lane environments increases the risk profile of the activity. In such cases, a site-specific TGS with either static work or a dynamic works convoy must be used.

The following process details the stages and steps that should be followed to install and remove a static lane closure on a motorway or other high-risk multi-lane environment. For clarity, the process, steps and diagrams have been simplified by only explaining one possible lane closure scenario, omitting some of the required signs and assuming that signs are not duplicated in the

median. Other scenarios such as a right lane closure, multiple lane closures or the duplication of signs in the median will require a modified process.

The four defined stages of the installation of TGS (in order of installation) are:

1. Stage 1 – Advance warning area
2. Stage 2 – Arrow board and pre-taper
3. Stage 3 – Taper and lane closure
4. Stage 4 – Lane closure and end of road work signs

Stage 1 – Advance warning area

The following steps must be undertaken for Stage 1 – Advance warning area as indicated by the circled numerals in Figure 6-5:

1. The advance warning vehicle must be positioned at least 2.0 m from the travel lane edge line. The advance warning vehicle must remain at a nominated location, with its VMS switched to Roadwork Ahead sign and work site speed zone sign to indicate commencement of the advance warning area.
2. The TMA must be positioned clear of the travel lane edge line, acting as a shadow vehicle for traffic control vehicle and workers on foot during sign and device installation. The TMA must be located no closer than 30 m from the installation activity.
3. The traffic control vehicle must be positioned no closer than 0.5 m from the travel lane edge line.
4. As the installation progresses, the TMA may leapfrog installed signs and devices provided there is adequate sight distance and gaps in traffic.
5. An additional, suitably sized vehicle must be in place if gaps in traffic are insufficient for the TMA to appropriately shadow the traffic control vehicle and workers on foot continually. Consideration should be given to using a second TMA as this additional vehicle.

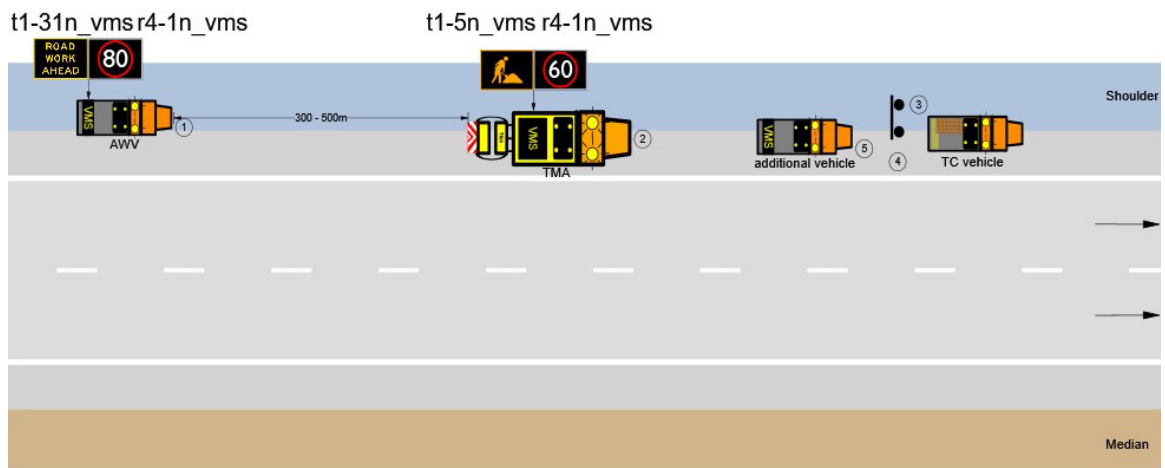


Figure 6–5 – Installation Stage 1: Advance warning area

Stage 2 – Arrow board and pre-taper

The following steps must be undertaken for Stage 2 – Arrow board and pre-taper as indicated by the circled numerals in Figure 6-6:

1. The advance warning vehicle must remain in its position at the commencement of the advance warning area* and be positioned at least 2.0 m from the travel lane edge line. As soon as the traffic control vehicle reaches the arrow board position, the advance warning vehicle must switch its VMS to the Roadwork 1km Ahead sign and work site speed zone sign.
2. The final Lane Status sign(s) must be located not closer than D from the commencement of the taper.
3. At *TMA position 1*, the TMA must shadow the traffic control vehicle and workers on foot as the arrow board sign is installed (this includes shadowing their exit from/entry to the traffic control vehicle).
4. The arrow board sign is installed and four cones at 12 m spacing, placed outside the edge line in advance of the taper.
5. At *TMA position 2*, the TMA must then move into the nearest travel lane to shadow the traffic control vehicle and workers on foot as they install the taper, from (6). See Section 6.4.1 for hierarchy of controls.

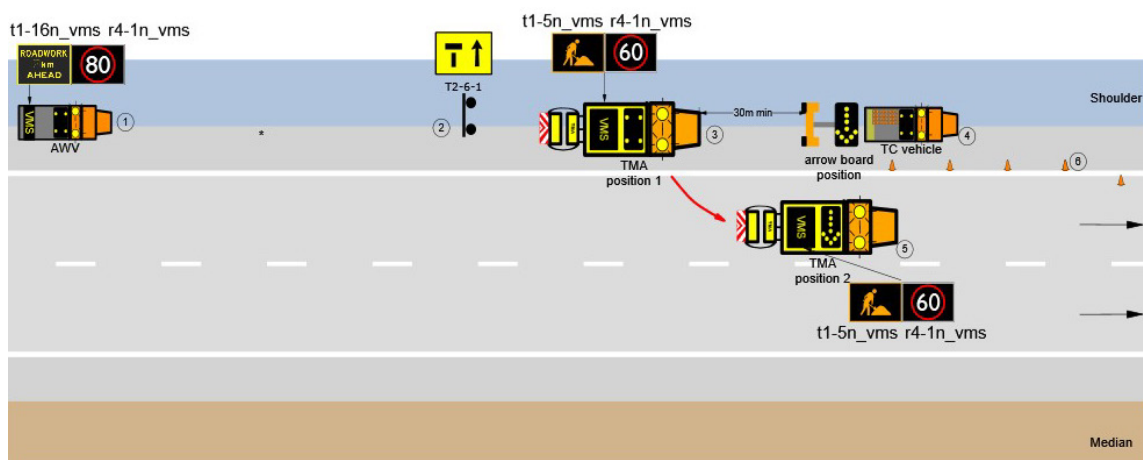


Figure 6–6 – Installation Stage 2: Arrow board and pre-taper

Stage 3 – Taper and lane closure

The following steps must be undertaken for Stage 3 – Taper and lane closure as indicated by the circled numerals in Figure 6-7:

1. The advance warning vehicle must remain in its position at the commencement of the advance warning area*, be positioned at least 2.0 m from the travel lane edge line and continue to display the Roadwork 1km Ahead sign and work site speed zone sign on its VMS.
2. The arrow board must clearly display the required merging movement.

3. At *TMA position 2*, the TMA must remain in the nearest travel lane to shadow the traffic control vehicle and workers on foot as the taper length is installed in accordance with Section 7.6.2.2, with the first cone (of the taper) placed 12 m from the last advance taper cone.
4. Appropriately spaced Hazard Warning Marker signs must also be installed.
5. After the taper is installed, the TMA must then move into *TMA position 3* via the shoulder.
6. The TMA must then remain as a shadow vehicle to the traffic control vehicle and workers on foot as the sign and device installation activity progresses in the partially closed lane. The TMA must not be located closer than 30 m from the installation activity.
7. The traffic control vehicle and workers on foot move along the partially closed lane and install delineation devices next to the lane line.

Note: For a closure of more than one lane and the installation of additional tapers, an additional arrow board and TMA must be used. Steps 2 to 6 above are repeated.

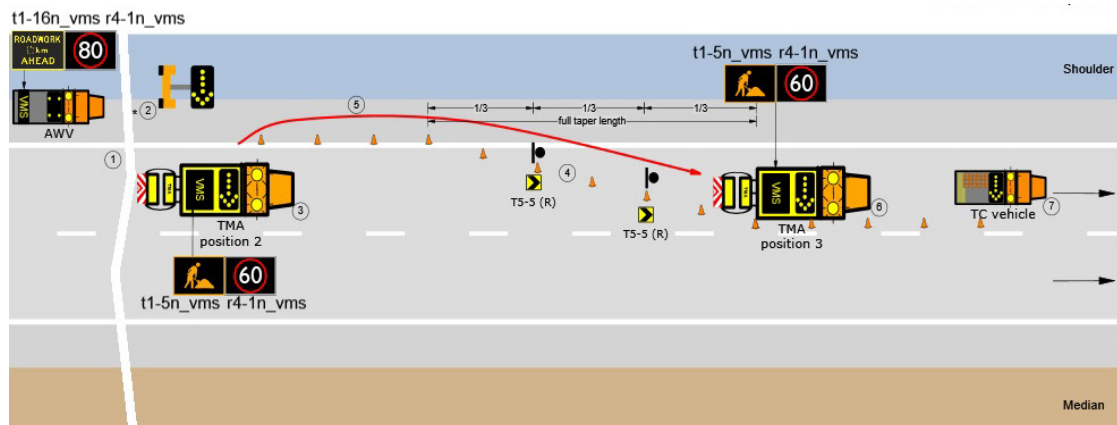


Figure 6–7 Installation Stage 3: Taper and lane closure

Stage 4 - Lane closure and end road work

The following steps must be undertaken for Stage 4 - Lane closure and end road works as indicated by the circled numerals in Figure 6-8:

1. The advance warning vehicle must remain in its position at the commencement of the advance warning area*, be positioned at least 2.0 m from the travel lane edge line, and continue to display the Roadwork 1km Ahead sign and work site speed zone sign on its VMS.
2. The arrow board must remain in its position at the start of the taper and clearly display the required merging movement. The taper length must be installed in accordance with Section 7.6.2.2.
3. To indicate the lane closure, barrier boards must be installed at a maximum spacing of 100 m.

4. The TMA must remain as a shadow vehicle to the traffic control vehicle and workers on foot as the sign and device installation activity progresses at (5). The TMA must not be located closer than 30 m from the installation activity.
5. The traffic control vehicle and workers on foot move along the partially closed lane and install delineation devices next to the lane line.
6. The End Roadwork sign must then be installed accompanied by the reinstated speed limit sign no further than D metres from the farthest worker on foot or plant item.
7. The TMA must remain as a shadow vehicle for the work area. The TMA must not be located closer than 30 m from the work area.

ITCP qualified person must then complete a drive around to confirm that the TGS is installed as designed.

The advance warning vehicle must remain in place for the duration of the road work operations and during the removal sequence.

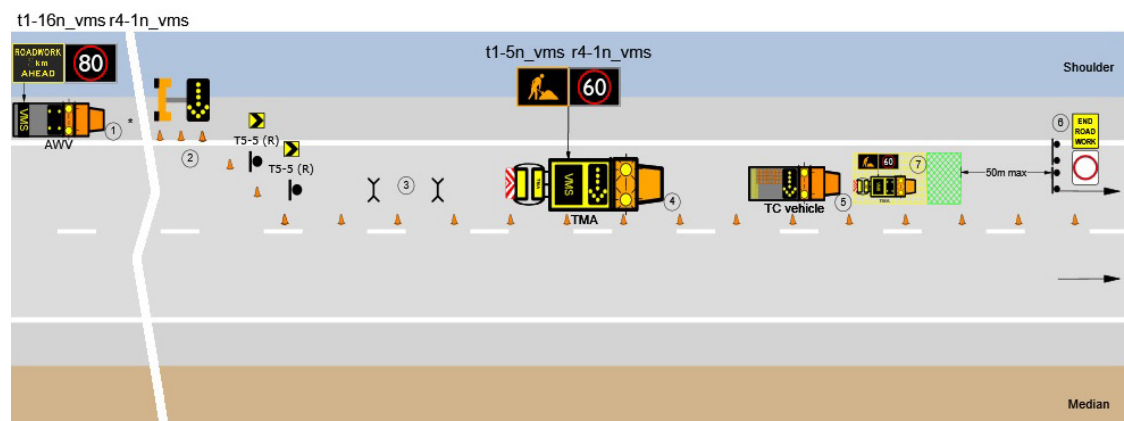


Figure 6-8 Installation Stage 4: Lane closure and end roadworks

Section 6.4.4 Removal

Section 6.4.4.1 General

When work is complete, the removal of signs and devices at the work site must be safely sequenced. Section 6.4.4.2 and Section 6.4.4.3 provide the general procedures for the removal of TTM signs and devices.

Section 6.4.4.2 Non-motorway and other non-high risk multi-lane environments

The following general requirements and recommendations apply to the removal of signs and devices on non-motorway and other non-high-risk multi-lane environments (such as two-lane, two-way roads or low risk multi-lane roads):

- Unless shown otherwise on the TGS or documented in the SWMS, the removal of devices should be undertaken in the reverse order of installation, progressing from the work area out toward the approaches.
- When removing delineation devices, such as cones, bollards or barrier boards used to close a lane:
 - a work vehicle must also be positioned between the workers and approaching traffic
 - an advanced warning vehicle should be used to warn road users of workers on foot
- A work vehicle must only proceed in a forward direction towards approaching traffic along the closed roadway if it is determined by the PWZTMP qualified person that it is safe to do so. This must not occur at night time where it may create motorist confusion or distraction, such as headlight glare.
- The removal of signs on central medians of multi-lane divided carriageways increases the risk profile of the activity. In such cases, a site-specific TGS with either static work or a dynamic works convoy must be used.

Section 6.4.4.3 Motorways and other high-risk multi-lane environments

The requirements in this section must be applied when removing signs and devices from any part of the roads listed in Table 6-20. These requirements should also be applied to roads not listed in Table 6-20, but meet all of the following high-risk multi-lane road criteria:

- multi-lane road
- posted speed limit greater than 85 km/h
- greater than 10,000 vpd.

As part of the risk assessment process, consideration should still be given to applying these requirements on other high speed multi-lane roads even though the road may not meet the above traffic volume criteria.

On motorways and other high-risk multi-lane environments, the removal of signs in reverse order (progressing out from the work site area toward the approaches) can be difficult due to higher speeds and distances between devices. As such, delineation devices must be removed in the reverse order back to the commencement of the taper. The signs in the advance warning area must be removed in the same order that they were installed.

The removal of the signs in the advance warning area must be carried out using the same method that was applied for their installation. An advance warning vehicle must be in place at an appropriate location and a TMA used to shadow the traffic control vehicle and the workers on foot.

The planning requirements and recommendations provided in Table 6-6 apply to the removal of signs and devices.

Table 6–6 Planning requirements and recommendations for removal

Item	Provision/s
Risk assessment	A site-specific risk assessment must consider the risks involved in each stage of the removal process and identify the controls necessary to eliminate or minimise those risks.
TGS and documentation	A site-specific process for the removal of signs and devices must be shown on the static works TGS or on a supplementary TGS by providing a numbered sequence for removal of the signs and devices. A SWMS or other generic document that applies to any location does not meet the above requirement for documenting how the static works TGS will be removed.
TGS design and approval	Both the primary TGS and (where used) the supplementary TGS for how it will be removed must be approved and verified in accordance with Section 7.9.
Co-ordination	There must be full co-ordination of any ITS infrastructure that may assist with the TGS removal, and this must be documented on the TGS. For example, this may involve the placement of advance warning messages on permanent variable message signs and/or a reduction of the speed limit where variable speed limit signs (VSLS) are provided.

The operational requirements and recommendations provided in Table 6-7 apply to the removal of signs and devices.

Table 6–7 Operational requirements and recommendations for removal

Item	Provision/s
Advance warning vehicle	At least one advanced warning vehicle must precede the TMA (acting as shadow vehicle). The advance warning vehicle must be fitted with a VMS that is sized in accordance with Section 6.5.3. During the removal of taper, or when the TMA is otherwise occupying an open trafficable lane, the advance warning vehicle should regularly communicate with heavy vehicle drivers via UHF radio and provide verbal advance warning of the TTM operation. The advanced warning vehicle must be in the road shoulder, offset from the travel lane edge line by at least 2.0 m. Offsets from the travel lane edge line of less than 2.0 m may be used, provided an additional TMA is used as the advance warning vehicle.
Shadow vehicle	A TMA must be used to shadow the traffic control vehicle and workers on foot. A TMA must not be located in the travel lane, unless: - the TMA is protecting TTM personnel who are in the process of removing signs and devices of the taper - the TMA is located inside the lane closure (behind the taper) - the removal activity is carried out using a dynamic (mobile) works convoy in accordance with Section 7.8.4.4.

Item	Provision/s
Number of vehicles	Additional, suitably sized vehicles must be in place if gaps in traffic are insufficient for the TMA to appropriately shadow the traffic control vehicle and workers on foot continually. For multi-lane closures involving more than one taper, two TMAs should be considered.

The removal of signs on central medians on motorways and other high risk multi-lane environments increases the risk profile of the activity. In such cases, a site-specific TGS with either static work or a dynamic work convoy must be considered.

The following process details the stages and steps that must be taken to remove a lane closure on a motorway or other high-risk multi-lane environments. For the purpose of simplicity and clarity of the process, it does not detail all the signs and devices to be removed and assumes that signs have not been duplicated in the median. Other scenarios such as a right lane closure or multiple lane closures will require a modified process.

The following are four defined stages removal of the TGS (in order of removal):

1. Stage 1 – Lane closure
2. Stage 2 – Taper area
3. Stage 3 – Advance warning area
4. Stage 4 – Arrow board.

Stage 1 – Lane closure

The following steps must be undertaken as indicated by the circled numerals in Figure 6-9:

1. The advance warning vehicle must be positioned at least 2.0 m from the travel lane edge line. The advance warning vehicle must remain at a nominated location, with the VMS switched to the Roadwork 1km Ahead sign and work site speed zone sign to indicate commencement of the advance warning area.
2. Relevant advanced warning signage must still be installed. The arrow board must be positioned such that it clearly displays the required merging movement, and the relevant taper length must still be installed.
3. Before removal commences, all workers and plant must be cleared from the work area.
4. The TMA must then raise the attenuator and reverse to an agreed point within the lane closure area. The attenuator must then be lowered, as such becoming the shadow vehicle. The traffic control vehicle and workers on foot may then remove relevant signs and devices within the lane closure area beginning with the End Roadwork and speed reinstatement signs and moving backwards. This step is repeated until the TMA is positioned at the commencement of the lane closure. During this process, the TMA must be no closer than 30 m from the removal activity.

5. When the TMA reaches the end of taper and all signs and devices in advance of the taper are removed, the TMA and traffic control vehicle must be then repositioned to the advance warning area. For efficiency a second TMA may be employed.
6. Having been repositioned and with the TMA now positioned in the lane in front of the taper, the traffic control vehicle and workers on foot are ready to commence removal of the taper.

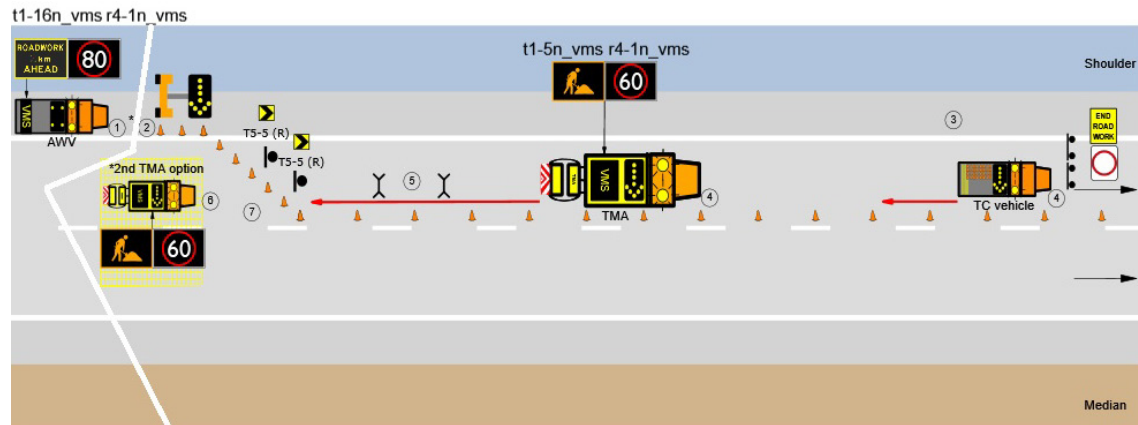


Figure 6–9 Removal stage 1: Lane closure

Stage 2 – Taper area

The following steps must be undertaken as indicated by the circled numerals in Figure 6-10:

1. The advance warning vehicle must be positioned at least 2.0 m from the travel lane edge line. The advance warning vehicle must remain at a nominated location, with the VMS switched to Roadwork 1km Ahead sign and work site speed zone sign to indicate commencement of the advance warning area.
2. Relevant advanced warning signage must still be installed. The arrow board must still be positioned such that it clearly displays the required merging movement, and the relevant taper length must still be installed.
3. With the TMA now positioned in the lane, in front of the taper, the traffic control vehicle and workers on foot may commence removal of the taper in reverse order from (5). The TMA must be no closer than 30 m from the removal activity.
4. An additional, suitably sized vehicle may be employed to provide additional cover on the shoulder.
5. The traffic control vehicle and workers on foot move in the reverse direction to remove the taper delineation devices.
6. After the taper has been removed, the TMA and traffic control vehicle must then be repositioned to the start of the advance warning area to commence the removal of the advance warning signage.

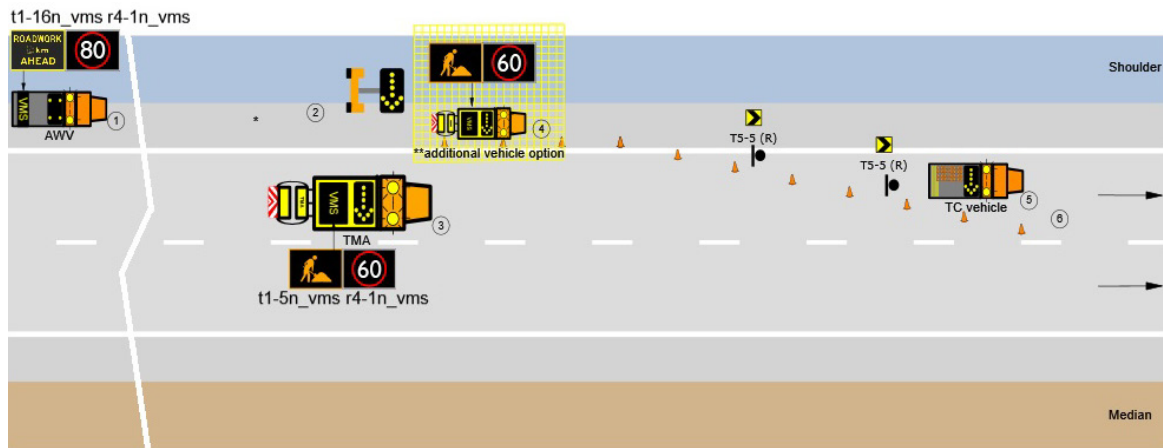


Figure 6–10 Removal stage 2: Taper area

Stage 3 – Advance warning area

The following steps must be undertaken as indicated by the circled numerals in Figure 6-11:

1. The advance warning vehicle must be positioned at least 2.0 m from the travel lane edge line. The advance warning vehicle must remain in advance of the TGS advance warning area and display on the VMS the Roadwork Ahead sign and work site speed zone sign.
2. The TMA must be positioned clear of the travel lane edge line, acting as a shadow vehicle for traffic control vehicle and workers on foot during removal of advance warning signs and devices. The TMA must be no closer than 30 m from the removal activity.
3. Beginning with the first sign/device after the advance warning vehicle, the traffic control vehicle and workers on foot may then remove subsequent signs and devices in a forward direction. The traffic control vehicle must remain positioned no closer than 0.5 m from the travel lane edge line.
4. The advance warning vehicle and TMA may then progress along the shoulder with the traffic control vehicle during the removal process.

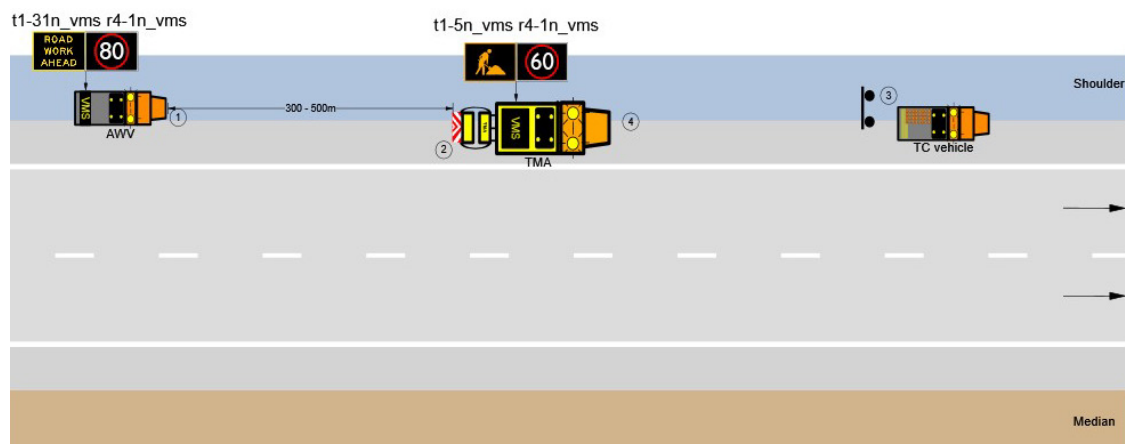


Figure 6–11 Removal stage 3: Advance warning area

Stage 4 - Advance warning area and arrow board

The following steps must be undertaken as indicated by the circled numerals in Figure 6-12:

1. The advance warning vehicle must be positioned at least 2.0 m from the travel lane edge line, with the VMS displaying the Roadwork Ahead sign and work site speed zone sign. The advance warning vehicle moves along with the TMA and traffic control vehicle during the removal process.
2. The TMA must be positioned clear of the travel lane edge line, acting as a shadow vehicle for workers on foot during removal of arrow board and cones. The TMA must be no closer than 30 m from the removal activity.
3. The traffic control vehicle must be positioned at least 0.5 m clear of the edge line and remove the arrow board and three remaining cones in advance of the taper.
4. All vehicles must then accelerate in shoulder and re-join traffic, de-activate warning beacons once they reach a speed of not less than 20 km/h below the posted speed limit.

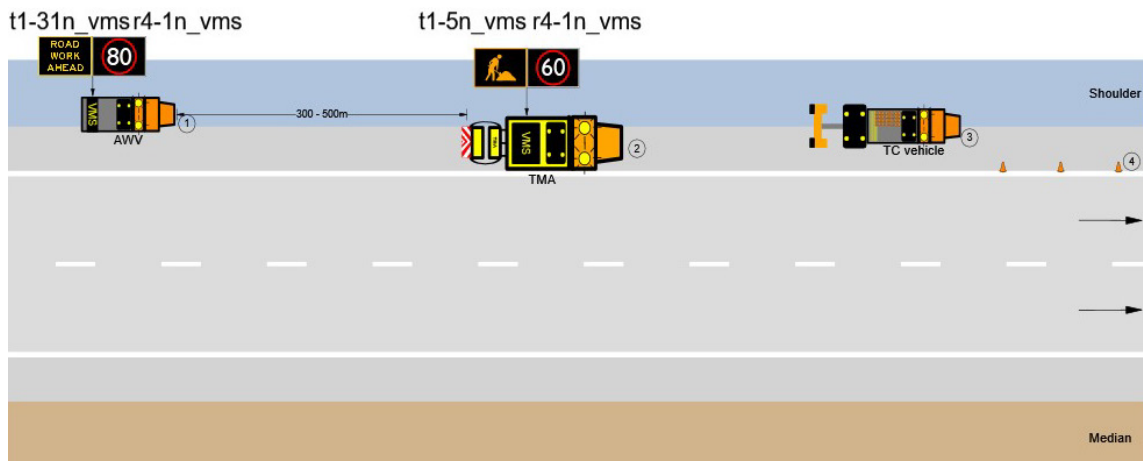


Figure 6–12 Removal stage 4: Advance warning area and arrow board

Section 6.5.2 Types of signs

Replace Table 6-5 in its entirety with the following:

Table 6-5 Examples of signs for a typical category

Category	Example signs	Sign number
Road hazard approaches and departure	TRAFFIC HAZARD AHEAD END TRAFFIC HAZARD	T1-10-1n T1-10-2n
Work site approaches and departures	ROADWORK AHEAD GRADER AHEAD END ROADWORK	T1-1 T1-4 T2-16

Category	Example signs	Sign number
Regulatory control of traffic	Speed limit ROADWORK PREPARE TO STOP STOP HERE ON RED SIGNAL	R4-212n T1-18 R6-6
Detours	DETOUR One lane each way symbolic Arrow marker symbolic	T5-1 T2-24 T5-6
To indicate road conditions	SLIPPERY symbolic SOFT EDGES ROUGH SURFACE POTHOLE GRAVEL ROAD Loose stones symbolic LOOSE SURFACE NEW WORK NO LINES MARKED	T3-3 T3-6 T3-7 T3-219n T3-13 T3-9 T3-14 T3-11
Lane and road closures	Lane closed symbolic ROAD CLOSED	T2-6-1 T2-4
Pedestrian control signs	PEDESTRIANS USE OTHER FOOTPATH LOOK BOTH WAYS TWO WAY TRAFFIC	T8-2 T8-3 T8-5
Vehicle height and mass restriction signs	LOW CLEARANCE X.Xm BRIDGE LOAD LIMIT Xt GROSS	R6-11 R6-3

Section 6.5.9 Requirements for specific signs

Replace Table 6–11 in its entirety with the following:

Table 6–11 Requirements and conditions of use for specific TTM signs

Sign	Conditions	Notes
ROADWORK AHEAD (T1–1) or (T1-31) BRIDGEWORK AHEAD (T1-2)	Must be used at: - Long-term road or bridge work sites. - A diversion of traffic along a side track or detour. - Unexpected conditions, such as loose stones or the absence of linemarking.	The signs may be used: - With the NEXT 2km (T1-28) sign for frequently changing work areas. - Short-term works where additional advance warning is warranted.

Sign	Conditions	Notes
ROADWORK X km AHEAD (T1-16) BRIDGEWORK X km AHEAD (T1-29)	Must be used where: - The approach speed is greater than 85 km/h. - Relevant sight distance is less than 150 m (to work area/end of queue etc).	This sign to be positioned X km from: - The start of the taper area. - The traffic diversion. - The traffic control position.
ROADWORK ON SIDE ROAD (T1-25)	Must be used in advance of an intersection to warn of the relevant activities on the side road where there is insufficient distance on that road to provide the required warning.	
SIDE ROAD CLOSED (T1-32)	Must be used in advance of an intersection where the side road is closed to all traffic.	
TRAFFIC HAZARD AHEAD (T1-10-1n)	Must be used to provide advance warning of a traffic hazard where: - The approach speed is greater than 65 km/h. - Relevant sight distance is less than 150 m (to the hazard). This sign must be placed a minimum of distance D from the start of the traffic hazard.	
END TRAFFIC HAZARD (T1-10-2n)	Must be: - Used in conjunction with the TRAFFIC HAZARD AHEAD (T1-10-1n) sign. - Placed at a distance D from the end of the hazard to indicate that normal traffic conditions have resumed.	Signs reinstating the existing permanent speed limit should be placed adjacent to this sign.
STOP HERE ON RED SIGNAL (R6-6)	Must be used: - Where traffic is required to stop in compliance with a PTS (placed 6 m in advance of the PTS). - With the SIGNALS AHEAD (T1-30 or W3-3) sign.	

Sign	Conditions	Notes
END ROADWORK (T2-16) or (T2-17)	Must be: - Placed at a distance D from the work site to indicate that normal traffic conditions have resumed when ROADWORK AHEAD (T1-1) or ROADWORK X KM AHEAD (T1-16) signs are used - Placed adjacent to or after any signs indicating the reinstatement of an existing permanent speed limit.	The T2-16 sign is preferred wherever space is available as the site allows it to be used.
SLIPPERY SYMBOLIC (T3-3) SOFT EDGES (T3-6) ROUGH SURFACE (T3-7) POTHoles (T3-219n) GRAVEL ROAD (T3-13) LOOSE STONES SYMBOLIC (T3-9) LOOSE SURFACE (T3-14)	Must be installed to warn motorists of conditions which make a roadway surface temporarily hazardous.	On long work sites these signs should be repeated at intervals of not more than 500 m.
TRAFFIC HAZARD (T1-10)	Must be used only when a more relevant sign is not available.	
REDUCE SPEED (T1-278n)	Must be used in conjunction with one of the above road conditions signs or another sign describing the hazard.	
NO LINES DO NOT OVERTAKE UNLESS SAFE (T3-12) NO LINES DO NOT OVERTAKE (TM3-12-1n)	Must be used in a two-lane, two-way road when: - Lines have been removed. - A new seal has been installed.	- T3-12 may be used where overtaking would normally be permitted in an oncoming traffic lane. - TM3-12-1n must be used where barrier lines would normally be installed and overtaking is not permitted.
NEW WORK NO LINES MARKED (T3-11)	Must be used on a multilane road when: - Lines have been removed. - A new seal has been installed.	Must not be used where there is an unacceptable risk of collision due to oncoming traffic. In such cases, T3-12 or TM3-12n as applicable must be used.

Section 6.8.7 Temporary pavement markings and markers

Replace the contents of Section 6.4 in their entirety with the following:

Pavement markings on temporary roadways and detours must be of a similar standard to that in use at either end of the adjoining sections of road. Where the adjoining road is delineated with edge lines, temporary roadworks must be similarly marked with an edge line.

Pavement markings and markers used at temporary work sites generally comprise:

- barrier, lane and edge lines
- turning arrows
- raised retroreflective pavement markers (temporary or permanent)
- projected pavement arrows (that is, pavement arrow images projected onto pavement surface by a road image projector).

Where, during or at the conclusion of pavement-surfacing works, a section of roadway is to be left for a period of time without linemarking, temporary raised pavement markers should be used to provide delineation at the dividing or lane lines. Application of the pavement markings must take place as soon as practicable.

Where temporary linemarking, that is, linemarking not in its final location, is required on the final wearing surface, or adjacent pavement, pavement marking tape should be used. Where used, pavement marking tape must meet the performance requirements of QA Specification R145.

Where it is determined that any temporary pavement marking or marker has become ineffective, remarking and/or replacement must be undertaken as soon as practicable.

Where a single carriageway is opened adjacent to, or used in lieu of, an existing dual carriageway length, pavement arrows (in tape if they are required to be removed from a final wearing surface) indicating the direction of flow of traffic must be placed as directed with the maximum allowable spacing being 500 m. The arrows must be removed if the section is then reincorporated as dual carriageway.

Where appropriate as part of static night works, 'projected pavement arrows' may be used to supplement existing traffic control measures, such as in advance of a merge taper or truck mounted attenuator (TMA). Projected pavement arrows must be dimensionally representative of those within AS 1742.2 with no distortion when projected. The use of projected pavement arrows must be documented as part of the TMP and the relevant TGS by a PWZTMP qualified person.

Where existing pavement markings are required after the temporary works, these may be masked over with suitable black tapes where they exhibit similar characteristics to the existing pavement colour and surface finish, during the works period. This may be a suitable consideration where limited traffic impact is expected, to limit the chance of dislodgement by

traffic. Otherwise, permanent removal must be undertaken and agreement must be received from the asset owner in accordance with Section **Error! Reference source not found..**

On long-term works raised retroreflective pavement markers complying with QA Specification R142 may be used in conjunction with temporary pavement markings. The spacing and application must be as specified in QA Specification R142.

Pavement markings and retroreflective markers should be considered in conjunction with the placement of other delineation devices and must be used where temporary safety barriers are used, to ensure road users are safely directed through the site without conflicting messages.

With the exception of projected pavement arrows, all pavement markings must be retroreflective in accordance with QA Specification R145. Where projected pavement arrows are used, illumination must be such that the projected image is clearly visible to approaching road users.

Section 7.6.2.2 Tapers

Replace caption of Table 7–4 with the following:

Minimum spacing between tapers

Add the following new section after Section 7.7.5

Section 7.7.6 Signposting traffic hazards

Plans must be in place to ensure that drivers are alerted to hazards that have been identified on the road carriageway, such as pavement damage or failure, prior to road works commencing to remediate the hazard. Until works can be undertaken to remediate the hazard, signposting and/or delineation can be used to mitigate the risk to road users. This signposting and/or delineation should alert road users to the hazard and clearly communicate the nature of the hazard and the actions that road users should take.

Signposting a traffic hazard does not remove or reduce the requirement to rectify the hazard as soon as practicable and should only be implemented until such time that works can take place to remediate the hazard.

An appropriate TGS must be implemented as soon as practicable after identifying the hazard until the traffic management is established for the road works. The TGS may involve advance warning signage prior to the hazard, sign posting at the hazard and other mitigation measures such as a speed limit reduction and/ or delineation devices to guide traffic around, past or through the hazard.

Given that traffic hazards can occur at any time, generic TGSs should be developed so that an appropriate TGS can be implemented immediately when responding to hazards arising from traffic incidents, natural disasters or other unplanned events.

Any generic or site-specific TGS for managing a traffic hazard must be designed and implemented in accordance with *Section 7.0 Traffic guidance schemes* and the following applicable provisions (see Figure 7–15a for an example layout):

- The TGS must include details of how traffic and if applicable other road users (that is, pedestrians, cyclists and so on) will be managed around, past or through the hazard site at all times until the traffic management is implemented for the road works that will repair the damage or remove the hazard. This includes the removal or covering of any signs that may no longer be applicable, particularly permanent speed limits.
- Unless the risk assessment of the hazard site indicates otherwise, the speed limit should remain unchanged if safe or, if not appropriate, should be no more than 20 km/h below the existing permanent speed limit, having regard to the identified site conditions (for example, pavement condition, damaged safety barrier), expected weather conditions and road characteristics such as traffic volumes, heavy vehicles, urban versus rural road environment.
- Inspections of the site should be undertaken in accordance with *Section 8.1 Work site inspections, reviews, and audits*.

As part of preparing the site to be left unattended, the actions shown in Table 7–6a should be taken.

Table 7–6a Considerations and actions

Considerations	Actions
Size of site	Minimise the length of the site as much as possible by providing minimum spacings to signage before and after the hazard(s) unless determined otherwise by the risk assessment.
Temporary speed limits	If the risk assessment has determined that a reduced speed limit should be installed, the design should consider minimising the length of the reduced speed zone (any changes to a speed limit must be approved).
Unsafe conditions and sign selection	Identify the hazard, such as loose material on the road surface, road erosion, a depression or hollow in the road surface and select the appropriate sign to warn road users accordingly.
Signs	Implement appropriate advance warning and road conditions signs – see Table 6-5 for sign options and spacing. TTM signs need to remain in place and serviceable while the site is not attended. Check that all signs are ballasted for windy conditions, reasonably level, mounted at the correct height and positioned correctly.
Delineation	Check that all delineation devices are clean, positioned correctly and adequately ballasted, for example, double weighed bollards.

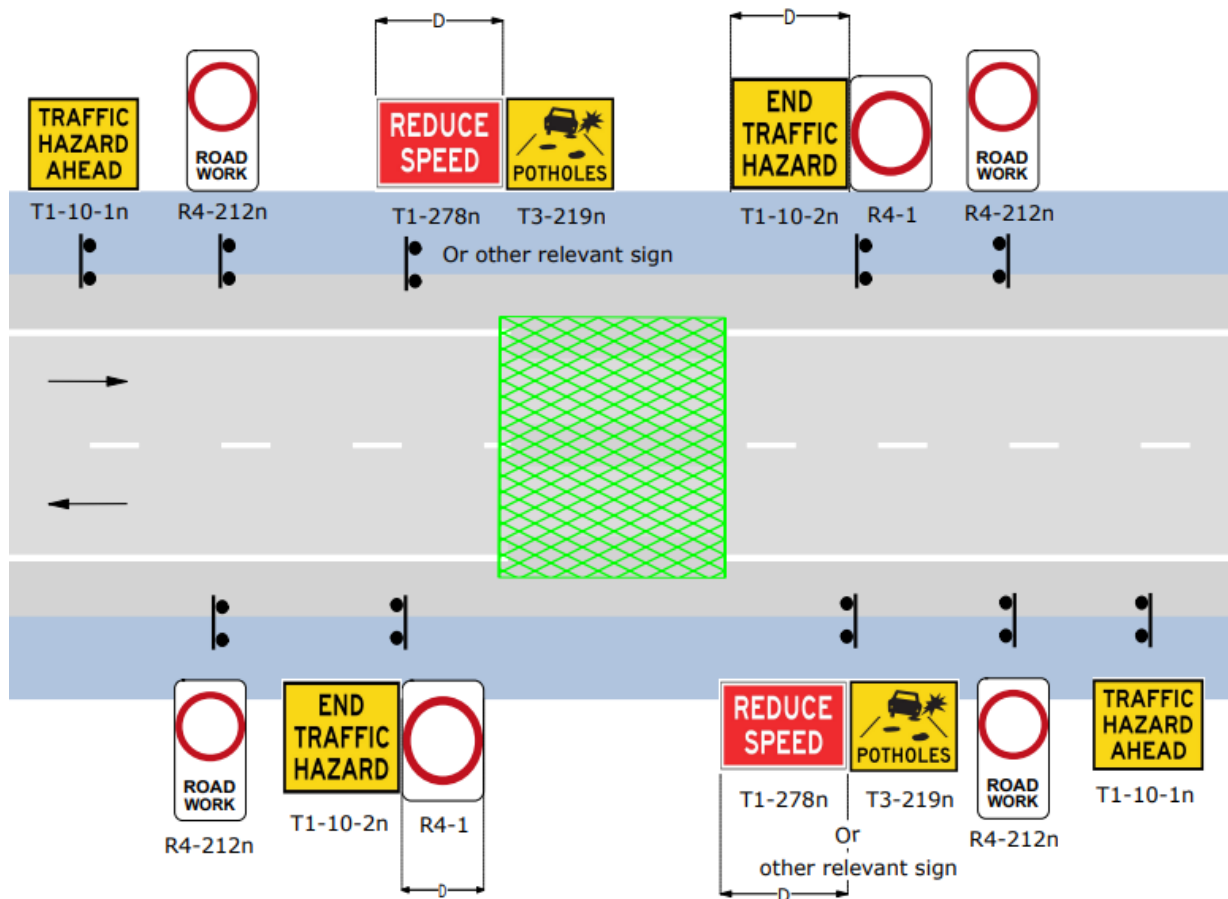


Figure 7–15a Example layout – sign posting a traffic hazard

Section 7.9 TGS approval and verification

Replace the contents of Section 7.9 in their entirety with the following:

Section 7.9.1 TGS approval

After the PWZTMP qualified person has designed or modified a TGS, the TGS must be reviewed and approved for use by a second PWZTMP qualified person, who is not the designer or modifier. The purpose of the approval is to ensure that the TGS:

- Has been designed in response to the risks and hazards identified in the relevant TMP, risk assessment and associated TTM specific documentation.
- Has been designed in accordance with the requirements of this document.
- Contains the relevant information for the ITCP qualified person to safely implement onsite including additional support vehicle/s and sequencing.

Any departures to the requirements of this document, including non-standard signs and devices proposed by the TGS must be appropriately approved in accordance with Section 2.8.

Additionally, the use of a manual traffic controller as part of the TGS must be undertaken in accordance with Section 5.4.

The TGS must include separate sign-off sections for the designer and approver. The sign-off sections must include the relevant persons:

- Full name
- Role
- Division/organisation
- SafeWork NSW card number
- Signature
- Date.

Section 7.9.2 TGS verification

After a TGS is selected or designed and approved, it must be verified as being appropriate for use at the work site, via the completion of the TGS verification.

A TGS verification must be undertaken to confirm the selected or designed TGS is fit for purpose. A TGS verification must be completed in accordance with Section 8.1.2 by an ITCP or PWZTMP qualified person. TGS verification must include an inspection of the work site where the TGS will be implemented.

Authorisation:

Approved by	Director Corridor Infrastructure and Engineering Asset Management Safety, Environment and Regulation
--------------------	--



For queries regarding this document
standards@transport.nsw.gov.au
www.transport.nsw.gov.au

Technical Direction – TD 00031:2022

Issue date: 04 November 2022

Effective date: 04 November 2022

Title: Traffic control at work sites

This technical direction is issued by the Asset Management Branch (AMB) as an update to *Traffic control at work sites*, Issue 6.1.

The update includes an amendment to G9-79 sign in Table 6-12.

1 Amendment to *Traffic control at work sites*, Issue 6.1

Section 6.5.10 in *Traffic control at work sites* is amended as follows:

Section 6.5.10 Signposting of roadwork speed zones

Replace only the Conditions column for Speed limit AHEAD (G9-79) sign in Table 6-12 as follows:

Table 6-12 Requirements and conditions of use of specific signs

Sign	Conditions
Speed limit AHEAD (G9-79) sign	Speed Limit AHEAD signs must be erected where the speed of traffic on the approach to the temporary speed zone is 35 km/h or more than the temporary limit. The Speed Limit AHEAD signs must be located 2D in advance of the initial roadworks speed zone (R4-212n) signs.

Authorisation:

Approved by	Director Roads & Traffic Engineering Technical Services Infrastructure and Place
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Transport for NSW

Traffic control at work sites

Technical Manual



Trial undertaken on Newell Highway, Peak Hill, NSW

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About this release

Title:	Traffic control at work sites
Document Number:	20.346
Author:	Senior Temporary Works Interface Manager, Technical Services, Infrastructure & Place
Authorised by:	Director Corridor Infrastructure & Engineering, Asset Management, Safety, Environment & Regulation

Issue	Date	Revision description
6.0	14 September 2020	Manual Update
6.1	28 February 2022	Minor Revision

Foreword

This Technical Manual has been developed by Transport and must be applied to the following:

- Transport work sites requiring temporary traffic management (TTM); and
- Works involving temporary traffic management being undertaken for or on behalf of Transport (by contractors, local government and public utility bodies or similar).

This is the sixth issue of the Technical Manual and its release seeks to harmonise Transport practice with National practice where practicable.

This Technical Manual is to be read and used by personnel responsible for designing, implementing, operating and inspecting temporary traffic management at Transport work sites relating to construction or maintenance activities.

Where this Technical Manual refers to a Transport roadwork site, it includes any Transport construction or maintenance work that impacts on the road network, this could include rail or maritime infrastructure projects.

The information in this Technical Manual is provided to assist Transport in meeting its obligations under the *NSW Work Health and Safety Act 2011* (WHS Act). This Act places a positive duty of care on persons conducting a business or undertaking (PCBU) to ensure the health and safety at work, so far as is reasonably practicable (SFAIRP), of:

- Workers they engage or cause to be engaged; and
- Workers whose work activities they influence.

Transport must also ensure, SFAIRP, that the health and safety of other persons is not put at risk from work carried out as part of the conduct of their business or undertaking.

The *NSW Work Health and Safety Regulation 2017* (WHS Regulation) prescribes the risk management approach that duty holders of a PCBU must apply. This includes:

- Identifying all reasonably foreseeable hazards that could give rise to health and safety risks;
- Assessing these risks; and
- Managing the risks to health and safety by eliminating them, or, where this is not reasonably practicable, minimising them.

The WHS Regulation describes the hierarchy of control measures that duty holders must implement if it is not reasonably practicable to eliminate risks to health and safety.

The Technical Manual does not provide an exhaustive list of controls and does not displace Transport's duties under the *WHS Act* (2011) and *WHS Regulation* (2017).

Compliance with the requirements of this Technical Manual by itself is not sufficient to ensure satisfactory outcomes for traffic management. Transport expects that professional judgement be used by competent personnel when performing TTM to ensure such outcomes.

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Amendment Record

This amendment record references the changes made against Version 6.0 (dated 14 September 2020)

Version	Section	Sub Section	Amendment	Issued
6.0	NA	NA	Issue 6: New edition, completely rewritten. Descriptions of changes to previous versions are no longer relevant, and are not listed.	14 September 2020
6.1	Throughout	Throughout	Corrections, minor amendments and internal consistency changes.	28 February 2022
	2	2.4.2	Removal of the requirement for Transport for NSW-specific refresher training for temporary traffic management qualifications.	
	4	4.3.3	Revision of the concept of “clear zones” in line with the Austroads <i>Guide to Road Design</i> .	
	6	6.9.1.2	Inclusion of a set of static signs that are permitted for display on a variable message sign (VMS).	
	6	6.9.2	Inclusion of new guidance material on the use of variable speed limit signs (VSLS).	
	7	7.7.3.2	Additional requirements for closing a lane on motorways or other multi lane roads.	

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1 Introduction

1.1 Purpose and scope

This Technical Manual is to be read and used by personnel responsible for designing, implementing, operating, reviewing and inspecting temporary traffic management (TTM) at Transport construction or maintenance work sites.

Where this Technical Manual refers to a Transport roadwork site, it includes any Transport construction or maintenance work that impacts on the road network, this could include rail or maritime infrastructure projects.

In the context of TTM, the Technical Manual contains instructions for the following:

- Manage risks associated with TTM;
- Develop a traffic management plan (TMP);
- Design, select, obtain approval, record and store a traffic guidance scheme (TGS); and
- Undertake traffic management in a number of specific situations.

The purpose of this Technical Manual is to ensure that traffic control at Transport work sites continually strives for best practice. It is also intended to help personnel understand their obligations under the *Work Health and Safety Act 2011* and the *Work Health and Safety Regulation 2017*.

For works conducted by contract, this Technical Manual complements the following Transport specifications:

- G10 (Traffic Management);
- D&C G10 (Traffic Management);
- G10M (Traffic Management (Maintenance Works));
- G22 (WHS Construction Work); and
- D&C G22 (WHS Construction Work).

Appendices of this Technical Manual noted as *mandatory* are key to the application of the Technical Manual. Appendices noted as *informational* are optional, and have been provided to assist practitioners with application.

Variations to and departures from the requirements of this Technical Manual must be in accordance with the departure process described in [Section 2.8 Departures from this Technical Manual](#).

1.2 Exclusions, evaluation and review

1.2.1 Exclusions

The requirements of this Technical Manual do not apply to 'standard work activities' outside the scope of construction or maintenance of a Transport managed road. Activities considered to be 'standard work activities' and excluded from the requirements of this Technical Manual include, but are not limited to:

- Transport Traffic Emergency Patrol work;
- School Crossing Supervisors;
- Emergency service related work including the use of crossovers as permitted/intended by their provision;

- Inspections that do not require traffic to be managed, or where traffic is not affected by the inspection;
- Heavy vehicle inspection conducted by Compliance Operations Inspectors; and
- Speed camera certifications and maintenance activities.

Standard work activities such as those listed above should be managed in line with divisional requirements and through the development of localised risk assessment and procedures to manage the risk associated with these works.

Unless specifically determined by the relevant division, any procedures that are developed for these activities need not adhere to this Technical Manual or prepared by a TTM qualified person, however, this Technical Manual may be used as a point of reference.

Readers are advised to consult the Transport website for additional information, guidance and requirements for these specialist activities.

1.2.2 Evaluation and review

Implementation of this Technical Manual will be evaluated as part of Transport's program of audits.

A Transport Traffic Control at Work Sites committee exists to assist in the review of this Technical Manual on a regular basis. Factors taken into account in these reviews include the circumstances of incidents at work sites and the results of safety audits and inspections.

The committee is also responsible for:

- Developing strategies to improve safe systems of work; and
- Providing input into traffic management innovation and new technology.

Feedback on the Technical Manual is encouraged and can be forwarded to the Director, Roads and Traffic Engineering via standards@transport.nsw.gov.au. When new information becomes available and new techniques are developed, they will be assessed and if suitable, will be incorporated into this Technical Manual. Amendments to the Technical Manual will be published on the Transport website.

1.3 Terms and definitions

For the purposes of this Technical Manual the terms and definitions provided in *Table 1-1* apply.

Table 1-1. Terms and definitions

Term	Definition
85th percentile speed	speed at or below which 85% of vehicles are observed to travel under free flowing conditions past a nominated point
AADT (annual average daily traffic)	total traffic volume over the whole year, divided by the number of days in the year
ADT (average daily traffic)	total traffic volume during a stated period, divided by the number of days in that period
advance warning signs	roadwork warning signs which have a general message and used in advance of other roadwork signs with a more specific message

Term	Definition
advance warning vehicle	vehicle used in advance of dynamic works to provide advance warning of those works
alternate flow	see shuttle flow entry
approach speed	speed of traffic approaching the work site measured in km/h and may be the speed limit applying to the road
built-up area	In relation to a length of road, an area in which either is present for a distance of at least 500 m or, if the length of road is shorter than 500m, for the whole road: - buildings, not over 100m apart, on land next to the road - street lights not over 100m apart - with or without kerb and channel
carriageway	portion of a road or bridge devoted particularly to the use of vehicles, inclusive of shoulders and auxiliary lanes
competent person	any person who has, through a combination of training, qualification and experience, acquired knowledge and skills to enable that person to perform specified tasks
condition signs	temporary signs indicating the condition of the road surface through the work area
consequence	effect such as illness, injury, loss of life or property damage resulting from a hazardous event occurring or a risk being realised
containment fencing	physical barrier that provides separation between pedestrians or cyclists and the work area, or between separate work areas, but not so rigid as to become a hazard if struck by a vehicle
controlled area	area of a roadway where temporary traffic management is in place, i.e. between the first advance warning sign and where normal traffic arrangement resume
contraflow	when a carriageway of a divided road is closed to traffic and the traffic is transferred to the other carriageway which then operates as a two-way road
controller	device which regulates the order and duration of the displays of the signal lanterns
crossover	access point from one carriageway to the adjacent carriageway for emergency vehicles (e.g. police, ambulance, fire etc.)
date of publication	date Issue 6.1 of the Traffic Control at Work Sites Technical Manual is made available on the Transport for NSW website
delineation	general term for treatments which regulate, warn or provide tracking information and guidance to drivers (e.g. linemarking, raised pavement markers, traffic cones, bollards and post-mounted reflectors are delineation devices)
dimension D	distance expressed in metres for the positioning of TTM signs, devices and determining other TTM related factors
escort vehicle	vehicle used under the Heavy Vehicle National Law to escort an Oversize Overmass vehicle on the road network

Term	Definition
footpath	paved area in a footway
footway	public way largely reserved for the movement of pedestrians
hazard	situation, condition or source that has the potential to lead to negative consequences, harm or loss, but not the negative outcome itself
high risk pedestrian	subgroup of pedestrians at greater risk to hazard such as persons with a disability, the elderly and children
high speed road	road which is posted at speeds greater than 85 km/h
high volume road	road which has traffic volumes greater than 20,000 vpd
implement traffic control plan (ITCP)	SafeWork NSW qualification principally concerned with the competency of implementing a work zone traffic management plan (TMP)
intermittent work	work which is undertaken on travel lanes, in gaps in traffic, without obstructing traffic and without compromising the safety of road workers
lead vehicle	vehicle used at the head of a work convoy on two-way roads (e.g. to give advance warning of the works to traffic approaching from the opposite direction and/or to enable the driver to alert following road workers of any impending hazard)
long-term work	work requiring traffic control for longer than one work shift and where some form of traffic control remains when the site is left unattended
lookout person	any person whose sole responsibility is to watch out for and warn road workers of approaching traffic
may	indicates permission or an option, often accompanied by one or more conditions
modification	with reference to a TGS and the positioning of signs and devices, any change which exceeds a permitted tolerance
motorway	motorway is used but not defined in roads or road transport legislation. A motorway is a divided highway for through traffic with no access for traffic between interchanges and with grade separation at some interchanges. Certain activities (such as the carriage of dangerous goods) may be restricted or prohibited by legislation. Includes tollways.
must	indicates a mandatory requirement
open road area	roadside development less frequent than that specified for a built-up area
pedestrian movement plan (PMP)	diagram showing the allocated travel paths for road workers and/or pedestrians around, past or through a work site
pilot vehicle	vehicle used to guide traffic by controlling the desired path and to manage speed through the work site

Term	Definition
portable traffic control device (PTCD)	portable device that removes the need for manual traffic control or allows traffic controllers to perform their roles at a safe distance from traffic (e.g. type 2 (automatic) portable traffic signals, type 1 (manual) portable traffic signals and portable boom barriers)
portable traffic signal (PTS)	approved traffic signals that are transportable and not permanently installed
prepare work zone traffic management plan (PWZTMP)	SafeWork NSW qualification principally concerned with the competency of developing or modifying a work zone traffic management plan (TMP) and/or a traffic guidance scheme (TGS)
prescribed traffic control device	sign, signal, marking, structure or other device to direct or warn traffic on a road (or part of a road)
Regional Road	see Schedule of Classified Roads and State and Regional Roads publication. See also Portal for State Roads
risk	possibility of suffering harm or loss
road	defined in the Road Transport Act 2013 as 'an area that is open to or used by the public and is developed for, or has as one of its main uses, the driving of motor vehicles'. Roads include bridges and tunnels.
road designer	an engineering practitioner approved by Transport, with at least 5 years of recent experience in designing roads to Transport standards
road occupancy	consists of any activity likely to affect the operational efficiency of the road network (e.g. an activity that requires the road to be used in such a way as to affect traffic flow)
road occupancy licence (ROL)	permit which conditionally allows the holder to use or occupy a specified road space at approved times
road user	any driver, rider, passenger or pedestrian using the road
roadside	area between the reserve boundary and the nearest road shoulder
roadway	portion of the road devoted to the use of vehicles, inclusive of shoulders and any auxiliary lanes
roadwork site	works that are conducted on a road, including bridge or light rail works that impact a road
safety barrier	physical barrier designed to resist penetration by an errant vehicle and as far as practicable, to redirect errant vehicles back into the travelled path
shadow vehicle	vehicle which provides close up protection to the rear of road workers on foot
short-term work	work requiring traffic control taking less than or equal to one work shift and where road conditions are returned to normal when the shift ends

Term	Definition
should	indicates a non-mandatory recommendation
shuttle flow (aka alternate flow)	where a single lane is used alternately by both directions of traffic (e.g. where insufficient width is available for a lane to be provided in each direction)
sight distance	distance between the point at which an approaching driver first sees the whole of an object and the object itself (in the context of this Technical Manual the object is the traffic control sign or device)
so far as is reasonably practicable (SFAIRP)	see WHS Act, Section 18
State Road	see Schedule of Classified Roads and State and Regional Roads publication. See also Portal for State Roads
tail vehicle	vehicle used in a dynamic work convoy to provide advance warning of the works and protection
temporary traffic management (TTM)	the organisation, arrangement, guidance and control of both stationary and moving traffic, including pedestrians, cyclists and all types of vehicles, around a hazard or work site for the safety of both road workers and road users
traffic control	direction of traffic around, past or through a roadwork site, accident or other disruption
traffic controller	person holding a SafeWork NSW qualification principally concerned with the competency of directing traffic in accordance with a work zone traffic management plan
traffic gap	time interval between the detection of two successive vehicles in a traffic stream
traffic guidance scheme (TGS)	diagram showing signs and devices arranged to warn traffic and guide it around, past or, if necessary through a work site or temporary hazard Note to entry: <i>Traffic guidance scheme (TGS) was previously referred to as traffic control plan (TCP)</i>
traffic management plan (TMP)	document detailing work to be undertaken, identifying associated risks and the accepted control measures to reduce risks by describing its effect on the general area, especially its effect on public transport, cyclists, pedestrians, motorists and commercial operations with required approvals from road authorities
traffic management strategy	document containing work related conditions and constraints to assist the development of a traffic management plan
traffic staging plan	road design drawings that show how traffic passes safely around, past or through the work area during various stages of a roadwork project
travelled path	part of the roadway which is available to vehicles and consisting of one or more running lanes

Term	Definition
vehicle movement plan (VMP)	diagram showing the preferred travel paths for vehicles associated with a work site entering, leaving or crossing the through traffic stream
vehicles per day (vpd)	number of vehicles observed passing a point on a road in both directions in a 24 hour period
vehicles per hour (vph)	number of vehicles observed passing a point on a road in both directions in a given hour
vulnerable road user	subgroup of road user including pedestrians, cyclists, motorcyclists, scooter users
work area	area on the road or within the road reserve where the construction or maintenance work is being undertaken
work site	an area of road or road reserve which includes the work area or areas and any additional length of road required for traffic control such as signs and tapers
work vehicle	vehicle or item of plant which undertakes the work and supports the road workers on foot within the work area

1.4 References

The following documents, provided in *Table 1-2*, are referred to in the text in such a way that some or all of their content constitutes requirements, recommendations or informative elements of this document.

Table 1-2. References

Document	Publisher
<i>AS 1742.1, Manual of uniform traffic control devices, Part 1: General introduction and index of signs</i>	<i>Standards Australia</i>
<i>AS 1742.2, Manual of uniform traffic control devices, Part 2: Traffic control devices for general use</i>	<i>Standards Australia</i>
<i>AS 1742.3, Manual of uniform traffic control devices, Part 3: Traffic control for works on roads</i>	<i>Standards Australia</i>
<i>AS 1743, Road signs – Specifications</i>	<i>Standards Australia</i>
<i>AS 2700, Colour standards for general purposes</i>	<i>Standards Australia</i>
<i>AS 4852.2, Variable Message Signs, Part 2: Portable Signs</i>	<i>Standards Australia</i>
<i>AS/NZS 1906.1, Retroreflective materials and devices for road traffic control purposes, Part 1: Retro-reflective sheeting</i>	<i>Standards Australia</i>
<i>AS/NZS 1906.2, Retroreflective materials and devices for road traffic control purposes, Part 2: Retro-reflective devices (non-pavement application)</i>	<i>Standards Australia</i>
<i>AS/NZS 3845, Road safety barrier systems and devices</i>	<i>Standards Australia</i>
Delineation guide	<i>Transport for NSW</i>
<i>Guide to Road Design</i>	<i>Austroroads</i>

Document	Publisher
<i>Guide to Temporary Traffic Management</i>	<i>Austroads</i>
<i>Guide to Traffic Management</i>	<i>Austroads</i>
IC-QA-G10, Traffic Management	<i>Transport for NSW</i>
IC-DC-G10, Traffic Management	<i>Transport for NSW</i>
IC-QA-G10M, Traffic Management (Maintenance Works)	<i>Transport for NSW</i>
IC-QA-G22, Work Health and Safety (Construction Work)	<i>Transport for NSW</i>
IC-DC-G22, Work Health and Safety (Construction Work)	<i>Transport for NSW</i>
QA Specification R145, Pavement Marking (Performance Based)	<i>Transport for NSW</i>
QA Specification 3352, Fluorescent Plastic Traffic Cones	<i>Transport for NSW</i>
QA Specification 3385, Barrier Boards	<i>Transport for NSW</i>
QA Specification M209, Road Openings and Restorations	<i>Transport for NSW</i>
Schedule of Classified Roads and State and Regional Roads	<i>Transport for NSW</i>
Traffic Signs Register	<i>Transport for NSW</i>
TS200, Register of ITS Field Equipment	<i>Transport for NSW</i>
TSI-SP-049, Traffic Signal Systems	<i>Transport for NSW</i>
TSI-SP-059, Type 1 Portable Traffic Signals	<i>Transport for NSW</i>
TSI-SP-060, Illuminated Flashing Arrow Signs	<i>Transport for NSW</i>
TSI-SP-081, Type 1 Portable Traffic Signals with Boom Barrier	<i>Transport for NSW</i>

2 Temporary traffic management policies

2.1 General

Transport has a primary duty of care to ensure the health and safety of all persons at Transport work sites. This extends to its employees, its contractors and members of the public.

To ensure the safety of people at work sites, a high standard of temporary traffic management and traffic control around, past or through those work sites must be provided. A site-specific plan for the management of traffic can be developed by considering the conditions that may be encountered at each site.

The design, selection and implementation of traffic control measures detailed in this Technical Manual are based on AS 1742.3, *Manual of uniform traffic control devices*, Part 3: *Traffic control for works on roads* and the Austroads *Guide to Temporary Traffic Management* (AGTTM). If this Technical Manual does not contain advice on a particular aspect of traffic control then the latest version AS 1742.3 and the relevant Part of the AGTTM applies, if the required information is available.

This Section details the key policies that Transport applies to temporary traffic management on work sites.

2.2 Traffic management plan

The purpose of a TMP is to assist in providing a safe work environment while maintaining a safe and efficient journey for all road users.

In accordance with Section 3 Traffic management planning process, a traffic management plan (TMP) must be developed for all temporary traffic management (TTM) works on Transport work sites, including maintenance work sites.

A TMP may be developed that applies to either of the following, depending on the nature of work to be completed, its risk profile and location:

- A specific road corridor; or
- A project or activity.

A TMP must ensure that works are arranged such that:

- Road workers are able to work safely;
- Road users are able to travel around, past or through the work site safely;
- Road workers and road users are separated wherever possible; and
- It does not impact or cause delay to road users or, if not reasonably practicable, it is minimised.

The TMP templates provided in Appendix A – Traffic management planning templates and tools can be used to assist in the collection of data and inform the development of TMP.

2.3 Traffic guidance scheme

In accordance with Section 7 Traffic guidance schemes (TGS), a TGS must be developed where the road environment is temporarily changed and the road users may be impacted. A TGS is a detailed layout of temporary signs and devices that communicate the TTM arrangement to guide traffic around, past or through a work site or temporary hazard. For the purposes of this document, a TGS is classified as one of the following:

- Generic;
- Site Suitable; or

- Site Specific.

Each TGS must be checked against risks identified in the TMP to ensure that the means of controlling or reducing these risks are in place.

This Technical Manual contains example work site layouts as diagrams throughout and in [Appendix D – Work type layout examples](#).

Note: These diagrams outline the types of controls that may be applied in a particular situation and do not include all signs/devices required. The diagrams are intended only to assist in the design of TGSs.

2.4 Personnel

2.4.1 General

The development, design and implementation of TMPs and associated TGSs must be undertaken by authorised and competent persons who are fit for duty in accordance with this Section.

2.4.2 Training

Personnel undertaking TTM in NSW must hold a valid traffic control qualification relevant to the type of work they are undertaking.

As of 1 July 2020, traffic control training in NSW is prescribed under the WHS Regulation 2017 and is managed by SafeWork NSW. As a requirement of the WHS Regulation 2017, a person undertaking traffic control work must hold the relevant qualification for the work they are performing. The three qualifications outlined in the WHS Regulation 2017 are as follows:

- Traffic Control (TC);
- Implement Traffic Control Plans (ITCP); and
- Prepare Work Zone Traffic Management Plans (PWZTMP).

For a list of SafeWork NSW approved training providers see the [Service NSW Public Register of Training Providers](#).

In this Technical Manual, all references to the training certification titled Implement Traffic Control Plans refers to the qualification currently recognised in NSW for implementing traffic guidance schemes.

Table 2-1 provides a summary of the three traffic control training qualifications with requirements and restrictions for each of the qualifications.

As required by the WHS Regulation 2017, Transport as a PCBU must ensure that any persons undertaking traffic control work have either completed the Safework NSW training for that type of traffic control work in the preceding 2 years, or have been carrying out that type of traffic control work in preceding 2 years.

Table 2-1. Traffic control training qualifications, requirements and restrictions

Qualification	Requirements	Restrictions
Traffic Controller This qualification provides the necessary certification to control traffic with a prescribed traffic control device.	Persons holding this qualification are permitted or required to: - Stop or direct road users using a STOP/SLOW bat or other accepted traffic control device; - Maintain traffic incident reports; - Operate a 2-way radio; - Understand the TGSs for the site; - Check traffic control signs are installed in accordance with the relevant TGS; - Assess and respond to changes in the environment, e.g. traffic volumes, weather conditions, road conditions, WHS and operational requirements; and - Carry out risk assessments for personal safety.	Persons holding this qualification must not: - Select or adjust a site suitable TGS; - Implement a TGS; - Modify a TGS; or - Design a TGS.
Implement Traffic Control Plans This qualification allows for qualified personnel to set up and work with TGSs at a work site and complete safety inspections.	Persons holding this qualification are permitted to: - Set up, monitor, and close down traffic control devices according to nominated TGS; - Identify safety implications of traffic control at roadworks; - Check, clean and store equipment on completion of work and close down a TGS; - Select an approved TGS to suit site conditions, traffic volumes and work activities; - Make adjustments to an existing TGS within the tolerances specified in <u>Section 7.10.3 Tolerances on positioning of signs and devices</u>; - Conduct an onsite check of a TGS to identify risks and hazards; - Ensure spacing between signs and traffic control devices is in line with a TGS; - Maintain traffic incident reports; and - Monitor traffic controllers.	Persons holding this qualification must not: - Control traffic with a STOP/SLOW bat or other traffic control device; - Make adjustments to an existing TGS which exceeds the tolerances specified in <u>Section 7.10.3 Tolerances on positioning of signs and devices</u>; or - Design a TGS.
Prepare Work Zone Traffic Management Plan This qualification allows for qualified personnel to design and modify Traffic Management Plans (TMPs), Vehicle Movement Plans (VMPs) and traffic guidance schemes (TGSs).	Persons holding this qualification are permitted to: - Prepare a Work Zone TMP; - Collect all required information about a given roadwork project to enable the preparation of a TGS; - Design a TGS, based on risk assessment, statutory and regulatory requirements, standards, road authority requirements and project brief; - Select and modify a TGS based on risk assessment, statutory and regulatory requirements, standards, road authority requirements and project brief; - Determine the recommended spacing between signs and traffic control devices in line with standards, measure width of trafficable surface and calculate edge clearances to barriers, cones and clearance to work personnel;	Persons holding this qualification must not: - Control traffic with a STOP/SLOW bat or other traffic control device; or - Implement a TGS.

Qualification	Requirements	Restrictions
	- Undertake safety inspections/checks on the effectiveness of TMPs and TGSs; - Conduct an onsite check and inspection of the plan and to identify any hazards or risks; and - Seek approvals required for a TMP and TGS	

2.4.3 Fitness for duty

Workers performing TTM must be fit for duty when reporting for work and during working hours (including breaks and travel time). This includes being free from the adverse effect of prescribed, over-the-counter and alternative medication which might negatively affect the ability to perform duties and/or pose a risk to the safety of workers and/or others.

Alcohol or prohibited drugs must not be consumed on a Transport work site or workplace at any time. Refer to the Transport [Drugs and alcohol procedure](#) for more information. All workers when undertaking work for, or on behalf of Transport, must comply with this procedure at a minimum. If work is being undertaken on a site that is under the control of a Principle Contractor, the Drug and Alcohol requirements of that Principle Contractor prevail only if the standard exceeds that required by Transport.

Workers should inform their manager where there is reasonable suspicion that anyone working on a roadwork site may be under the influence of drugs or alcohol.

2.4.4 Personal Protective Equipment (PPE)

Workers performing TTM must wear approved high-visibility clothing, including wet weather clothing where appropriate, in accordance with Transport's current [Personal Protective Equipment Procedure](#). PPE must be clean, bright and not obscured by or covered with other clothing.

Additional PPE such as hearing, eye and foot protection must also be worn as required by the relevant Safe Work Method Statement (SWMS) and in accordance with Transport's current [Personal Protective Equipment Procedure](#).

2.5 Traffic control

Traffic control is any direction of traffic around, past or through a roadwork site, accident, hazard or other disruption. A summary of the key policies contained in [Section 5.4 Traffic control](#) are as follows:

- Traffic control must be used if road users are to be directed to deviate from a traffic regulation, such as crossing a barrier line;
- The implementation of traffic control must be conducted in line with the hierarchy of controls with the elimination of harm to workers and the travelling public considered in the first instance;
- Where traffic control is required, a portable traffic control device (PTCD) must be used rather than using a manual traffic controller when the existing permanent speed limit is greater than 45 km/h, see [Section 5.4.2 Traffic control types](#). [Section 5.4 Traffic control](#) provides the conditions under which a manual traffic controller may be used;
- Where PTCDs or traffic controllers are used, approach speeds of traffic must be reduced to less than 65 km/h; and
- All persons operating a portable traffic control device or performing manual traffic control must be:
 - Qualified with 'Traffic Control' training; and

- Authorised by the relevant road authority.

Section 5.4 Traffic control permits the use of a manual traffic controller provided all of the following conditions are met:

- The use of a PTCD is demonstrated to not achieve the safest outcome;
- The decision to use a manual traffic controller instead of a PTCD is documented; and
- Approval is granted by the one-up manager of the PWZTMP qualified person or the nominated divisional representative.

Additionally, a manual traffic controller may be used in instances of emergency response.

If a manual traffic controller has been justified and approved in the TMP, the manual traffic controller must have four (4) cones placed at 4 m spacing at a safe location immediately preceding the location of the traffic controller on the edge line, centre line or both, and the appropriate signage in accordance with Section 5.4 Traffic control.

2.6 Signs and devices

Signage must be installed in accordance with Section 6 Signs and devices. When using signs they must be:

- Placed before the roadwork begins and be removed as soon as they are no longer required;
- Regularly checked to ensure they are still relevant, in good mechanical condition, have not moved, rotated or blown over, are clean, not faded and have good night-time visibility as necessary; and
- Inspected to ensure they remain clearly visible and command attention to road users and are not obscured by vegetation, vehicles, plant or other signs and devices, and are displayed in the correct sequence.

2.7 Work sites

The work site is the length of road which includes the area where the work is being undertaken and any additional length of road used for traffic control including signs, tapers, traffic lights and other devices. The work site is made up of five smaller areas detailed in Section 7.6.2 Components of the work site.

A work area is a component of a work site and is occupied by workers, plant and materials. Work areas must be:

- Designed so that the minimum length and width of a road is closed at each stage to minimise disruption and inconvenience to road users while maintaining work site safety and efficiency;
- Staged to ensure minimum disruption to traffic especially at peak times, nights, weekends, holiday periods and during special events; and
- Monitored with action taken if lengths of traffic queues or delays occur which are greater than those predicted and allowed for.

2.8 Departures from this Technical Manual

2.8.1 General

It is acknowledged that during the planning or implementation of TTM, there might be instances where the mandatory, minimum requirements contained in this Technical Manual are not achievable, or are not achieving the required level of risk management. In these instances, a variation to a requirement or a

departure developed and approved in accordance with this Section, may provide a better outcome. The rationale for all such decisions must be documented.

There are three broad categories of departures shown in [Table 2-2](#). Departures must be managed according to the processes within the relevant category to allow for the effective management of risk.

None of the processes contained within this Section enable departures from Road Occupancy Licence (ROL) requirements.

Table 2-2. Departure categories

Departure	Examples	Section
Use of unapproved signs – includes variation to existing sign designs or the introduction of a sign that is not contained with the Traffic Signs Register	- Modification of an existing sign design within the Traffic Signs Register; or - Development of a new sign specific to site activity	Section 2.8.2 Use of unapproved signs
Use of unaccepted device – includes the introduction of a device or innovative process that is not already accepted for use via this Technical Manual	- Use of a new or varied device for controlling traffic; or - Use of new or varied to delineation method	Section 2.8.3 Use of unaccepted devices
General departures – refers to a variation to a mandatory requirement in this Technical Manual that does not fall into a 'sign' or 'device' category	- Variation to the approved minimum lane width; or - Variation to the approved minimum edge clearance/shoulder width	Section 2.8.4 General departures

The requirements of this Section apply to all TTM planning or implementation being undertaken for or on behalf of Transport, including works managed by a Principal Contractor or Industry Partner, in consultation with the relevant Transport representative for the project.

2.8.2 Use of unapproved signs

All signs used to direct or warn traffic at a roadwork site must be approved, designed and used in accordance with this Technical Manual and the Transport electronic [Traffic Signs Register](#).

When it is determined that a work site requires a new sign, or modification to a standard sign, an Innovative/Non Standard Sign Design Request must be completed and submitted to standards@transport.nsw.gov.au.

Design and approval of a non-standard sign are undertaken by the Guidance and Delineation team within Road Specialists.

If a sign design is determined to be of benefit to Transport on an ongoing basis, it may be considered for inclusion on the [Traffic Signs Register](#).

2.8.3 Use of unaccepted devices

Under the Road Transport Act 2013, a prescribed traffic control device must not be installed, displayed on, above or near a road without appropriate authority. Devices detailed in this Technical Manual and the

relevant specifications where required, provide the authorisation and conditions for use by which those devices may be installed on a Transport work site.

All traffic control devices, including road markings, traffic signals, or any other device used to direct or warn traffic at a roadwork site must be accepted for use in accordance with this Technical Manual or another Transport standard, Technical Manual or specification.

Any device that is not approved for use by Transport must not be used without the appropriate written authorisation as per the Roads Transport Act 2013.

Written authorisation for the conditional use of a device for the purposes of TTM may be granted by the Director, Roads and Traffic Engineering via an application to standards@transport.nsw.gov.au.

Once submitted, the Traffic Engineering team will review the application and make an assessment for use considering:

- Existing standards and approvals required;
- Alternative solutions available;
- Safety and operational benefits;
- Constraints and risks; and
- Justification for its use.

In some cases, for authorisation to be granted, a field trial may be requested prior to or during its use as described in the application.

Any such trial will be requested by the Director, Roads and Traffic Engineering in consultation with the applicant. The purpose of the trial is to determine the safety, efficacy, operational requirements and suitability of the device for its use in this and other similar applications. The outcomes of trials may result in the integration of the device into this Technical Manual or other relevant documentation.

2.8.4 General departures

Where a mandatory requirement of this Technical Manual cannot be achieved or does not achieve an acceptable level of risk management, the following process must be followed:

1. Following the framework of Section 3.3.4 Risk assessment, the PWZTMP qualified person must undertake a risk assessment which includes:
 - a. Description of the work where departure is required;
 - b. Mandatory requirement not being met with detailed description why it cannot be met;
 - c. Options investigated
 - d. Proposed variation to the requirement, including if the variation is:
 - Aligned with an accepted and existing national practice document, such as AS 1742.3 or the AGTTM;
 - Aligned with an accepted and published standard of another Australian road authority; or
 - A risk based departure with no alignment to another standard or practice.
 - e. Risks introduced as a result of the proposed variation;
 - f. Additional controls needed to manage the introduced risks; and
 - g. The residual risk after the proposed variation and mitigation measures are applied.

2. Based on the anticipated residual risk and proposed standard applied, the PWZTMP qualified person must seek approval for the variation from the relevant Transport representative. Where the residual risk is determined to be:
 - a. **Low to Medium** – a Transport representative with authority at least Delegation 5 or higher (4, 3, 2 or 1) is required to approve the variation;
 - b. **High** – a Transport representative with authority at least Delegation 4 or higher (3, 2 or 1) is required to approve the variation.
3. After approval, the PWZTMP qualified person must update the TMP and other relevant documents such as traffic staging drawings and TGS in accordance with Section 8.2 Record keeping of TTM documentation. The TMP must include:
 - a. A copy of the approved risk assessment;
 - b. A summary of the departure including:
 - i. Description of the work where departure is required;
 - ii. Mandatory requirement not being met;
 - iii. Detailed reason for the requirement not being met;
 - iv. Options investigated;
 - v. Approved variation;
 - c. Minimum controls needed to manage the introduced risks; and
 - d. Supporting information such as drawings and correspondence etc.

Throughout the process, consultation should be undertaken with relevant subject matter experts where required, such as Traffic Engineering, Road Design or other specialists. Any changes to the TMP must be approved in accordance with Section 3.3.6 TMP approval and review scheduling.

Appendix A – Traffic management planning templates and tools contains a template that may be used to capture this information for inclusion in the TMP.

3 Traffic management planning process

3.1 General

Temporary traffic management (TTM) is one of the highest risk activities on a roadwork site. As such, TTM planning, in accordance with this Technical Manual, must be undertaken to assist with meeting work health and safety requirements and to provide a safe work environment while maintaining road user safety and network efficiency.

Traffic management planning is a risk management process that requires the input of relevant stakeholders to eliminate or manage risk to workers and all road users so far as is reasonably practicable (SFAIRP). The purpose of TTM planning is to:

- Establish the context of the road network, work activity and work environment;
- Assess the risk to road workers and road users by identifying, analysing and evaluating all actual and potential risks;
- Develop a strategy for traffic management based on the context and risks, which determines if traffic is required to be directed 'around', 'past' or 'through' the work site or temporary hazard;
- Develop and implement controls specifically related to the traffic management strategy. This includes design of site specific TMP, traffic guidance scheme/s (TGS) and obtaining required approvals; and
- Establish effective monitoring and review processes to ensure ongoing effectiveness of the process.

The TTM planning process, shown in [Figure 3-1](#) involves the development of a:

- Traffic Management Strategy; and
- Traffic Management Plan (TMP).

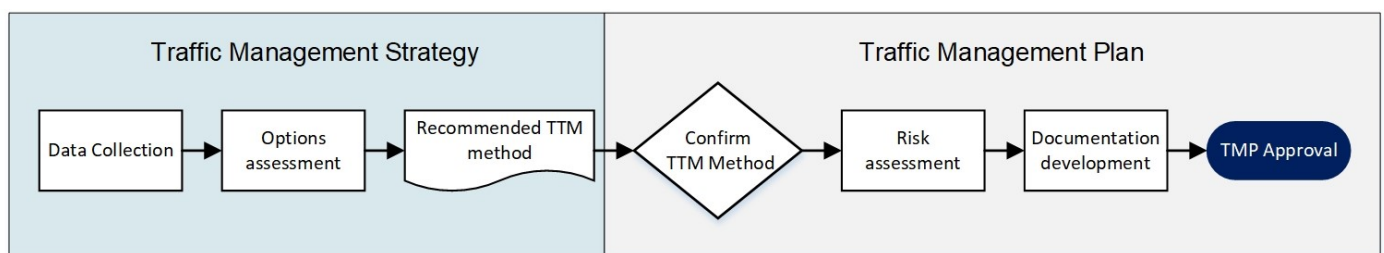


Figure 3-1. Overview of the traffic management planning process

A traffic management strategy and TMP must be developed for all roadworks. A TMP may be developed that applies to either of the following, depending on the nature of work to be completed, its risk profile and location:

- A specific road corridor; or
- A project or activity.

For example, for maintenance of a section of road, it might be more practical to develop a single traffic management strategy and TMP that is applicable to all maintenance work within a specified boundary.

Alternatively, for a project that involves a series of activities completed within a defined timeframe, a traffic management strategy and TMP for the project would be more appropriate.

An iterative process should be adopted in collaboration with relevant stakeholders to adopt the most appropriate traffic management approach and develop the associated documents for the work. For the purposes of Section 3:

- **Client** – refers to the person or team within Transport requesting or commissioning the works. The Client has the ability to access and provide relevant information about the work required and road environment. It does not strictly refer to the people funding the work.
- **Delivery partner** – refers to the team or business engaged to undertake the works. The delivery partner is responsible for making the final determination on the selected traffic management method and managing the development of site specific requirements such as a risk assessment and TGSs.

Table 3-1 provides some examples of client and delivery partner entities for different scenarios, however it is important to note that the client and delivery partner might be different for each activity.

Table 3-1. Example client and delivery partner entities

Example	Client	Delivery Partner
When maintenance works are completed internally	Transport Infrastructure Services	Transport Regional Maintenance
When maintenance works are completed externally	Transport Regional Maintenance Delivery	Principal Contractor
For road maintenance council contract (RMCC) works	Transport Delivery Strategy/Transport Infrastructure Services (or equivalent)	Local Council
For Transport Major Projects	Transport Regional Project Delivery	Principal Contractor
For Survey Works	Transport Major Project Office or Transport Regional Maintenance Delivery	Transport Engineering Services – Survey and contracted traffic control company

3.1.1 Traffic management method

A considered and consultative approach to traffic management planning allocates appropriate time and provides enough information to determine and apply the safest traffic management method. A traffic management strategy, completed early in the design safety lifecycle supports the appropriate allocation of time, funds and resources, and allows for consultation in determining the safest and most efficient way for road users to interact with the work site. In accordance with the hierarchy of controls, the three overarching TTM methods are described in **Table 3-2**.

Table 3-2. Traffic management methods

Traffic management method	Description	Examples
*Around (elimination)	An around method is where traffic is completely separated from the work area. An around method is the preferred TTM method where achievable, as a majority of risks associated with TTM are eliminated and it generally provides the lowest overall net risk option. This method must be considered as the first option, however if it cannot be achieved, justification must be provided in the TMP.	Examples of around methods include: • A road closure requiring a detour of all traffic. • Construction of a sidetrack. • Contraflow of traffic via a separated median.

Traffic management method	Description	Examples
Past (isolation or engineering)	A past method is where substitution, isolation and engineering controls are used to guide traffic along an adjacent path to the work area. A past method includes the use of a barrier or shifting of traffic to provide complete separation of workers and traffic.	Examples of past methods include: • Contraflow without a separated median. • A lateral shift taper. • Use of an accepted temporary barrier system.
Through (administration and PPE)	A through method relies on administrative, training and PPE controls only. A through method does not provide separation of traffic to the work area and requires the passage of traffic through the work area. A through method must only be considered when around and past strategies are not achievable or the risk generated by installing those options outweigh the safety benefit.	Examples of through method include: • Directing road users immediately over the work area. • Separation only achieved by use of cones or bollards. • Pilot vehicle used to platoon road users.

Note* to Table 3-2: It is acknowledged that the adoption of an 'around' traffic management method might require additional time, budget, area and community consultation. The earlier these are considered in project development, the greater the ability to incorporate into a TMP.

3.2 Developing a traffic management strategy

3.2.1 General

To effectively manage the risk associated with TTM, it is important that the conditions and constraints associated with the works are understood. When commissioning or ordering works to be completed, the client must ensure that sufficient resourcing, including time, funds and information have been provided to the delivery partner to enable effective planning of works.

For this to be achieved, the client must develop a traffic management strategy through the collection of relevant data and information and provide this to the delivery partner. A person developing the traffic management strategy does not require any formal TTM qualifications; however experience and knowledge in TTM, or an ability to consult those with the relevant experience is recommended. A traffic management strategy is comprised of three elements:

- Data collection;
- Options assessment; and
- Recommendation.

Data collection, further detailed in [Section 3.2.2](#) is the minimum mandatory information that must be provided by the client to the delivery partner within the traffic management strategy.

The 'options assessment' described in [Section 3.2.3](#) and the 'recommendation' described in [Section 3.2.4](#) should also be developed and provided by the client, however these are not mandatory. If they are not provided by the client, then they may be generated by the delivery partner during the development of the TMP.

Additionally, for more complex works, a draft TMP should be developed by the client to demonstrate feasibility of the works. During project development, the draft TMP should be provided to relevant stakeholders for consideration, such as survey and geotechnical, to support traffic management investigations.

In alignment with the [Transport Design safety lifecycle management framework](#), the information needed by the client is best collected and assessed early in the project cycle; ideally during the strategic phase, prior to the environmental impact statement or project boundaries being determined. This information should be completed and consulted as part of the Health and Safety in Design process to ensure the considerations of constructability are also factored into the options.

For works requested by a client outside of the Design safety lifecycle management framework, consultation with the relevant subject matter expert must be undertaken.

Appendix A – Traffic management planning templates and tools provides a template that may be used to facilitate development of a traffic management strategy.

3.2.2 Data collection

Data must be collected to inform the delivery partner about the way work will be undertaken and details of the work location. This information is fundamental to informing the TMP and the work specific risk assessment that is completed prior to the design of TGSs.

The client must provide the following information, where relevant, to the delivery partner in order to facilitate the development of the TMP:

Site related information

- Name of the project;
- Activity/work to be performed;
- Details of intersections impacted by the project length;
- Location of the work; and
- Details of the project phase for which the strategy is being developed.

Relevant site related data

- Cross section including description of unique features in cross section, and photo/aerial of the location of works;
- If the work location is in an urban or rural setting;
- Existing permanent and operating speed limits of all roads in work sites;
- Traffic volumes and composition in average daily traffic (ADT) and annual average daily traffic (AADT), if available, and AM and PM peak times;
- Details of heavy vehicle access requirements including vehicle types, e.g. restricted access vehicle (RAV) and over-size, over mass (OSOM) vehicles, and where available indication of length, mass, width, and heights of vehicles and percentage of types;
- Details of crash history;
- Details of intersections impacted by the project length; and
- Details of vulnerable road users and other facilities, including but not limited to:
 - On-street parking;
 - Transport facilities and infrastructure such as bus stops, train stations and tram stops;
 - Clearways;
 - Cycle ways; or
 - Footways.

Constraints

- Road environment constraints such as heritage considerations, utilities, cuttings, significant cut/fills, bridges, guardrails, limited shoulders and other environmental constraints like threatened species; and
- Details of significant traffic generators including their location, duration/time restrictions and other impacts. Examples of significant traffic generators are:
 - Local centres;
 - Retail, business, entertainment and community facilities;
 - Events;
 - Schools; and
 - Mines.

In addition to the above, any other relevant considerations should be provided to the delivery partner as part of the traffic management strategy.

3.2.3 Options assessment

After the work and site related information has been collected, the client should assess, detail and document the options available relating to the TTM methods detailed in [Table 3-2](#). The purpose of the options assessment is to provide the delivery partner with an understanding of the suitability and availability of ways in which traffic can be managed to enable the highest level of controls for road workers to be implemented. As such, when undertaking the options assessment, at least one *around* option should be considered and detailed.

It is acknowledged that information required to undertake an options assessment might not be readily accessible to the client, and in these instances, consultation should be undertaken with the delivery partner. The options assessment should provide detail of the multiple options assessed, and in particular, give information relating to both of the following traffic management methods:

- Around – the client should determine options available for around. It is recognised that there will be instances where an around option is not feasible, and in such cases, the reason for non-feasibility can be stated in the options assessment. When investigating suitability for around options the client should provide information to the delivery partner regarding:
 - A detailed description of the *around* route including any detours;
 - The duration that the *around* method will be in place, in either days, weeks or months;
 - The hours of operation of the *around* method, whether day and/or night and where known, the hours of operation;
 - Details of the benefits of the option in terms of cost and/or time;
 - Details of any constraints that will apply to the *around* method;
 - The estimated total cost to implement the *around* method, including route upgrades or infrastructure requirements;
 - Other considerations such as works traffic impacts, e.g. access and egress;
 - Stakeholder consultations required to ensure method is implemented successfully and that any affected residents or businesses have been notified; and
 - Approvals required to implement the *around* method.

- Past – the client should determine and provide information to the delivery partner regarding:
 - A description of the *past* method;
 - The anticipated duration that the *past* method will be in place, in either days, weeks or months;
 - The hours of operation of the *past* method, whether day and/or night and where known, the hours of operation;
 - The type of temporary traffic management that will be required to facilitate the *past* option;
 - Details of the benefits of the option in terms of cost and/or time;
 - Details of any constraints that will apply to the *past* method;
 - Other considerations such as works traffic impacts, e.g. access and egress;
 - Stakeholder consultations required to ensure method is implemented successfully;
 - Approvals required to implement the *past* method; and
 - Provision of clear access for emergency vehicles.

When developed by the client, the option of through does not need to be considered as this option is managed by the delivery partner. When developed, the options assessment must also be provided to the delivery partner.

3.2.4 Recommendation

After data collection as per Section 3.2.2 Data collection, and available methods for *around* and *past* traffic management have been identified in accordance with Section 3.2.3 Options assessment, a recommendation can then be made to the delivery partner on the preferred method.

Where a recommendation is made to the delivery partner, the recommendation should consist of:

- A preferred method;
- Note:** *A combination of two or more methods may be suitable.*
- A justification for the recommendation;
 - A description of the critical risks and other considerations that the delivery partner is to be aware of; and
 - Any additional comments.

Following the options assessment and where an *around* or *past* option is not recommended, the client then acknowledges that a *through* method is the only remaining alternative.

The final decision relating to *around*, *past* or *through* is the responsibility of the delivery partner and is determined following the provision of the above information with sufficient time for the TMP to be developed.

3.3 Developing a traffic management plan

3.3.1 General

A TMP shows how site specific works are integrated into the operation of the road network and additionally provides for the safety of workers and road users while maintaining the service provided by the road network. Management of work on roads must balance the need to protect road workers from traffic, while maintaining a safe and efficient road network for road users including vulnerable road users. The process of developing a TMP must be monitored and reviewed to ensure it is completed effectively.

All resources and documentation for a TMP must be stored in a records management system, to enable and support audit and review during the life of the project and after the project is complete. See [Section 8.2 Record keeping of TTM documentation](#).

After receiving all information needed to develop a TMP, including the traffic management strategy, the PWZTMP qualified person must develop and verify the TMP for the works. The TMP must contain at a minimum:

- A summary of information provided by the client in the form of a template provided by the client;
- An options assessment if not completed by the client in the traffic management strategy;
- Details of the traffic management method selected, including the decision and justification;
- A risk assessment for the works undertaken;
- Site specific documentation including relevant site specific TGS, Vehicle Movement Plans, Safe Work Method Statements (SWMS), Road Occupancy Licence and Speed Zone Authorisation; and
- Approvals, including formal acceptance of the TMP, TGSs and monitoring and review strategies.

[Appendix A – Traffic management planning templates and tools](#) provides a template that may be used to facilitate development of a traffic management plan.

3.3.2 Verification of traffic management strategy

As there can be a significant amount of time between the completion of the traffic management strategy and development of the TMP, verification of the information provided in the traffic management strategy must be undertaken. This includes:

- Verifying if all relevant information is provided and if not, the person developing the TMP must obtain the information from the client representative prior to completing the TMP;
- Verifying the accuracy of all previously collated data;
- Identifying any other work or site considerations that may have arisen since the completion of the traffic management strategy;
- Listing the details of any changes which impact the options identified in the traffic management strategy; and
- Listing additional options that may be available, and not identified in the traffic management strategy.

3.3.3 Decision of TTM method

After the information described in [Section 3.2 Developing a traffic management strategy](#) has been provided by the client and a verification of the information has been completed, the delivery partner must determine the TTM method, or combination of methods, required to safely perform the work or activity. When deciding the TTM method the following should be considered:

- *Detour options:*
 - Does the length and route of the detours proposed by the client introduce a disproportionate amount of disruption to the road users?
 - Does the detour introduce unacceptable impact/s on surrounding areas or unacceptable delays, resulting in undesired road user behaviour?
 - Is the cost of upgrading the route or intersections prohibitive when compared to the project value and duration?

- **Site location:**
 - Does the site of the works contain curves, crests, vegetation, existing signage or infrastructure that may obstruct signs and devices needed for certain strategies?
- **Work area:**
 - Does the area needed to safely perform the work justify the full closure of sections of road?
- **Vulnerable road users:**
 - Do desire lines of pedestrians, cyclists, motorcyclists and users of scooters impact on works or create undesired interaction between these road users and traffic?
- **Community facilities and needs:**
 - Does the presence of schools, hospitals, retail outlets, public transport routes or other facilities create conflict with the work?

After considering the above factors, a decision regarding the TTM method must be made. If the method that is selected differs from that recommended by the client, the reasons for the departure to this recommendation must be documented in the TMP and advice provided back to the client. The feedback to the client is critical in ensuring the client is informed of site specific considerations for future reference.

The decision to place traffic around, past or through must be documented with a justification of the decision made in the TMP.

3.3.4 Risk assessment

3.3.4.1 General

After a TTM method has been determined, the delivery partner must then identify the hazards and risks associated with the works through the development of a risk assessment. The purpose of this risk assessment is to determine the controls required for the protection of the road workers and road users.

A risk assessment is generally completed in accordance with the divisional requirements; however, it must comply with the requirements of the Transport procedure for [WHS Risk Management \(PN066P02\)](#) at a minimum. The information and data collected during development of the TTM strategy will need to be considered.

3.3.4.2 Hazard and risk identification

The types of information that should be reviewed and engagement activities that may assist in preparing and identifying hazards includes:

- An examination of the site layout (ideally a site visit should be undertaken);
- Road network maps, including heavy vehicle route maps;
- Traffic volume viewer;
- Consultation with workers and other relevant stakeholders;
- Risk registers, including risk registers from work sites that perform similar activities;
- Operational requirements and constraints; and
- Australian Standards and other Codes of Practice.

Hazards and risks that may be identified include, but are not limited to those listed in [Table 3-3](#)

Table 3-3. Hazards and risks

Hazards and risks	
Moving traffic	Traffic generating special events
Queued traffic	Non-compliance with temporary speed limits
High volume traffic	Reduced lane and shoulder widths
High vulnerable road user activity	Compromised access points
Other construction activity or roadworks in close proximity to proposed work site	Emergency vehicle access
Rising and setting of the sun	Overhead power lines or other utilities
Traffic speed and compliance behaviour	Horizontal (curves) and vertical (crests/sags) alignment
Traffic composition	Crash history
Number and location of traffic control points	Site vehicle access and egress points
Exposure and proximity of workers to live traffic	Topographical constraints
Length of delays for road users	

3.3.4.3 Risk analysis

After the hazards have been identified, a risk analysis must be carried out using the divisional framework or framework provided in [Figure 3-2](#) and [Table 3-4](#). The risk analysis is used to identify the likelihood of harm or damage caused by exposure to the hazard and the consequences of that risk occurring. This analysis must be based on all standard treatments for those risks being in place, with their established degree of effectiveness.

[Figure 3-2](#) provides the framework for the analysis of risk in the development of a TMP.

Risk evaluation matrix								
Risk ratings: Very high VH High H Medium M Low L			Consequence					
			Insignificant C6	Minor C5	Moderate C4	Major C3	Severe C2	Catastrophic C1
Likelihood	Almost certain	L1	M	H	H	VH	VH	VH
	Very likely	L2	M	M	H	H	VH	VH
	Likely	L3	L	M	M	H	H	VH
	Unlikely	L4	L	L	M	M	H	H
	Very unlikely	L5	L	L	L	M	M	H
	Almost unprecedented	L6	L	L	L	L	M	M

Figure 3-2. Risk evaluation matrix

[Table 3-4](#) provides the consequence and likelihood measures for the risk analysis.

Table 3-4. Likelihood and consequence measures

Consequence measures		Likelihood measures	
Rating	Description	Rating	Description
Insignificant	Illness, first aid or injury not requiring medical treatment. No lost time	Almost certain	- Expected to occur multiple times (10 or more times) during any given year - Expected to occur at least 1 in every 4 times the event or action occurs (more than 25% chance of occurrence) - This risk is known to occur frequently.
Minor	Minor injury or illness requiring medical treatment No lost time post medical treatment	Very likely	- Expected to occur occasionally (1 to 10 times) during any given year. - Expected to occur between 1 in 4 and 1 in 10 times the event or action occurs (10 to 25% chance of occurrence). - This risk is known to occur often.
Moderate	Minor injuries or illnesses resulting in lost time	Likely	- Expected to occur once during any given year. - Expected to occur between 1 in 10 and 1 in 100 times the event or action occurs (1 to 10% chance of occurrence). - This risk is known to have occurred on occasions.
Major	1 to 10 serious injuries or illnesses* resulting in lost time or potential permanent impairment	Unlikely	- Expected to occur once every 1 to 10 years. - Expected to occur between 1 in 100 and 1 in 1,000 times the event or action occurs (0.1 to 1.0% chance of occurrence). - This risk could occur but not often.
Severe	Single fatality and/or 11 to 20 serious injuries or illnesses* resulting in lost time or potential permanent impairment	Very unlikely	- Expected to occur once every 10 to 100 years. - Expected to occur between 1 in 1,000 and 1 in 10,000 times the event or action occurs (0.01 to 0.1% chance of occurrence). - It is unusual that this risk occurs but it has happened.
Catastrophic	Multiple fatalities and/or more than 20 serious injuries or illnesses* resulting in lost time or potential permanent impairment	Almost unprecedented	- Not expected to occur in the next 100 years (less than once every 100 years). - Expected to occur less than 1 in 10,000 times (if ever) the event or action occurs (less than 0.01% chance of occurrence). - Any risk can occur but it is very improbable that this risk will occur within the large number of events.

Note* to Table 3-4: serious injury or illness is defined by the WHS Act section 36

For additional information relating to analysing risk as part of a risk assessment, refer to the Transport procedure for [WHS Risk Management \(PN066P02\)](#).

3.3.4.4 Risk control

In order to decide the controls that are reasonably practicable, the person developing the TMP must consider the following principles:

- Severity of the risk/s mitigated by that option;
- Knowledge of that risk;
- Perceived benefit of the option;
- Nature of the works;
- Practicality of the option proposed; and
- Cost of removing or mitigating that risk.

All possible control measures must be considered, and a decision made on the control measures that are most effective and reasonably practicable.

The controls selected must prioritise the safety of workers and road users, while minimising the delay to traffic and costs associated with treatments.

For each of the risks identified, the [WHS Risk Management](#) process must be applied to ensure risks are managed using the hierarchy of controls framework.

The factors that must be considered in the selection of controls are provided in [Table 3-5](#).

Table 3-5. Factors for selection of temporary traffic controls

Key factor	Explanation
Application of minimum Standards	The cumulative impact of applying multiple minimum standards and the potential for an unsatisfactory outcome to occur as a result of the combination.
Contingency planning	Determine the way traffic will be managed in the instance that work cannot be completed as planned. This is particularly important for work that is not fully protected by temporary safety barriers. Contingency measures that can be implemented immediately should be employed.
Method and location of traffic control	Determine the way traffic will be controlled and the locations control is required. Possible control methods include: • Type 2 (automatic) Portable Traffic Signal (PTS) – fixed time operation (no traffic controller); • Temporary PTS (Type 1) – manual operation (traffic controller positioned remotely); • Boom barriers – manual operation (traffic controller positioned remotely); or • STOP/SLOW bat – manual operation (traffic controller exposed to traffic). Where traffic control has been identified as necessary, it must meet the requirements of Section 5.4 Traffic control and the relevant approvals must be sought for the use of a manual traffic controller.
Speed management	• Only credible speed limits should be implemented that will be self-enforcing. • Reduced speed zones must not be used in the place of more effective means of traffic control, but to complement such controls.

Key factor	Explanation
	- The reduced speed zone must only be used when work is being undertaken or when otherwise needed. 40 km/h speed zones should only be used when workers are on foot and the site is active. 30 km/h (see Section 4.5 Speed zones). - Engage with NSW Police Traffic and Highway Patrol Command to determine if enforcement activities will be required to assist with compliance.
Aftercare provisions	- Provision for night work traffic controls must be made where works will be undertaken during hours of darkness. - Roadwork sites, whether in operation or shutdown during the hours of darkness, must continue to be managed appropriately.
Emergency access	Provision for emergency services access through a roadworks site to another location to access the site in the event of an emergency on site during the works must be made.
End-of-queue management strategy	Where applicable, control measures must be implemented to stop or reduce the speed of traffic and result in either a static queue or slow moving queue (i.e. 20 km/h below the normal travel speed) in accordance with Section 4.6 End-of-queue management .

3.3.5 Site-specific documentation

The site-specific documentation that must be developed after finalisation of the TMP and risk assessment includes:

- A TGS which shows the placement of all temporary signs and devices. TGSs must be clear and show all traffic controls for each stage of the works;
- Road occupancy licences;
- Plans showing access to local properties and side roads affected by temporary works;
- WHS documentation including SWMSs and strategies for dealing with unplanned events or incidents;
- Emergency Management Plan to manage incidents within the traffic controlled area (vehicle break down, vehicle fires, pedestrians hit, car crash etc.);
- Approved list of TTM personnel and contacts;
- Vehicle Movement Plans showing mandated travel paths for vehicles to enter, leave or cross through traffic;
- Traffic incident plan; and
- Design drawings for temporary roadways and detour upgrades where applicable to the TTM strategy.

Other documentation that may be developed or included within the TMP includes:

- Speed zone authorisations;
- Traffic staging arrangements including Traffic Staging Plans and the time periods during which staging will be in operation;
- Traffic switches, which are often high risk activities that must have a detailed program of critical activities involved during the time of the switch;

- Pedestrian and cyclist movement plans showing allocated travel paths around, past or through the work site or temporary hazard. This should include provision for safe and unhindered access to public transport services; and
- Records of consultation with public transport operators.

3.3.6 TMP approval and review scheduling

The final step of developing a TMP must detail the approval, monitoring and review requirements of the TMP as provided in [Table 3-6](#).

Table 3-6. TMP approval, monitoring and review

Key factor	Requirements
Approval	The TMP must be reviewed and endorsed by the designer's one-up manager or other nominated person as determined by the divisional requirements. The one-up manager or nominated person must hold a current Prepare Works Zone Traffic Management Plan qualification. This approved TMP must be used to inform the development of all TGSs for the work as described in Section 7 Traffic guidance schemes (TGS) .
Monitoring	Monitoring and review is important throughout the TMP process to ensure that the TMP remains current and addresses all risks at the work site for the duration of the project or activity. To ensure that the TMP is kept up to date, details of the process to facilitate review and continuous improvement must be detailed in the TMP. This may include audits and inspections as per Section 8 Work site inspections, recording and reporting .
Review	After the TMP has been implemented, TTM reviews must be undertaken in accordance with Section 8.1 Work site inspections, reviews and audits . Other TMP reviews must be completed as required by the G10 Specification, where applicable. Where G10 Specification is not applicable, a TMP review must occur at least every 12 months or where any on-site or works changes occur. All relevant changes must be considered and recorded in the TMP with any changes made by an appropriately qualified person. A copy of all documentation relating to the endorsement of the changes must be available to be accessed, either electronically or in hard copy, by the person responsible for the works.

4 Inputs to a traffic management plan

4.1 General

When developing a traffic management plan, the following principles must be taken into account to ensure the safety of road workers and road users:

- Type of work being performed;
- Temporary works design principles;
- Providing for specific road users;
- Speed zones;
- End-of-queue management; and
- Credibility of work sites.

Detailed considerations of these principles when developing a TMP are provided in the following Sections.

4.2 Types of work

4.2.1 General

In this Technical Manual, work is classified by duration and activity. The terms used in this Technical Manual to classify work by duration are short-term work and long-term work. The terms used in this Technical Manual to classify work by activity are static work and dynamic work.

Table 4-1 provides a summary of the permitted relationships between static, dynamic, short and long term work.

Table 4-1. Relationships between work types

	Short term	Long term
Static work	✓	✓
Dynamic work	✓	✗

4.2.2 Short-term work

Short-term work applies to traffic management when work does not exceed the duration of a single shift and the work site is continuously attended. Examples include activities such as maintenance of freeway safety barriers, linemarking or installation of treatments such as raised pavement markers or audio tactile linemarking.

With all short-term work, roadway conditions must be returned to normal, without traffic control or after-care provisions, when work has been completed.

Short term work must be performed as either static work or dynamic work.

4.2.3 Long-term work

Long-term work applies when work is performed over a duration greater than one shift and traffic management is used between shifts. For long-term work, a traffic guidance scheme might need to operate both day and night with the work site left unattended. Long-term work must be arranged and undertaken in accordance with the relevant requirements of static work.

Long term work must only be performed as static work.

4.2.4 Static work

Static work is classified as work that is completed at a fixed site for a period of time with TTM. Static work sites may involve complex traffic arrangements and are often established so that a site can be left unattended during or between work shifts. The use of a static work site must be a risk based decision, where the risk of setting up static controls is considered against the protection provided by those controls.

Static work might include the use of lane closures, roadwork speed zones, changes to linemarking, detours, shoulder closures etc. and often with some form of traffic control.

The length of a static work site may range from 100 m to many kilometres of work area. If traffic is to be stopped for any duration at a static work site, then the length of the work site should be considered in the TMP in relation to risk (particularly end-of-queue).

Static work must be performed as either short-term work or long-term work. For information on designing a TGS for static work see [Section 7.7 Designing a TGS: Static work](#).

4.2.5 Dynamic work

Dynamic work is classified as work that is short in duration and moves along a length of roadway. Dynamic work is classified into three sub-categories:

- *Frequently changing work* – regularly moves between successive locations, either in or outside of a traffic lane where minimal warning is required to advise road users of the presence of workers;
- *Continuous work (previously known as mobile work)* – progressively moving in vehicles along the roadway; or
- *Intermittent work* – work which is undertaken on travel lanes, in gaps in traffic, and requires no adjustment that affects road users on the roadway.

Dynamic work may be completed:

- In a lane, on an edge line or on a shoulder; or
- On a verge, median or footpath.

Depending on the activity, dynamic work may involve:

- Workers only;
- Plant only; or
- A combination of both.

See [Section 7.8 Designing a TGS: Dynamic work](#) for examples and further details of dynamic work.

Dynamic work must be performed as short-term work only. For information on designing a TGS for dynamic work see [Section 7.8 Designing a TGS: Dynamic work](#).

4.2.6 Risk Management through work type selection

In all instances, the work type that enables the highest order control through the hierarchy of controls must be selected. Most often for long term works, this is the selection of a static site protected by a detour or safety barriers. It is recognised however that the selection and set up of static methods for short term work can in itself generate an increased level of risk (e.g. by exposing workers to traffic for longer periods during the setup of the site than the time required to perform the work). In these instances, a dynamic work method may be selected, however the worker protection must always be prioritised in accordance with the hierarchy of controls framework shown in [Figure 4-1](#).

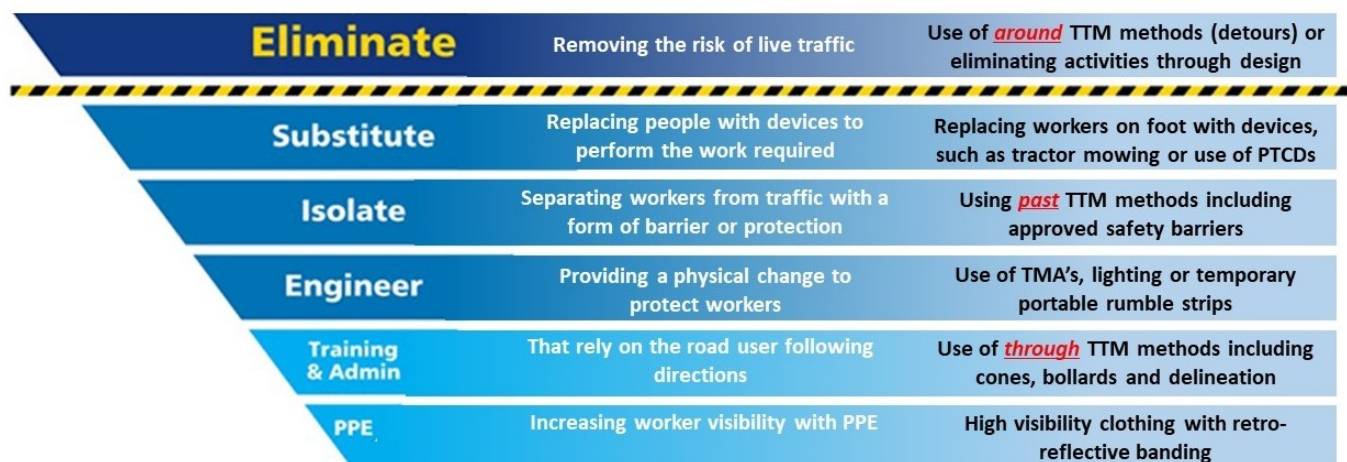


Figure 4-1. Hierarchy of controls framework

4.3 Temporary works design principles

4.3.1 General

When designing a TMP, appropriate temporary design principles must be used to ensure a safe road environment for road workers and road users. In some instances, these principles will align with the permanent road design standards.

4.3.2 Lane widths

The minimum width for lanes carrying traffic around, past or through a work site must be in accordance with [Table 4-2](#). Lane widths should also consider accommodating the turn paths of large vehicles expected to negotiate the work site.

Table 4-2. Minimum lane widths

Speed of traffic (km/h)	Minimum lane width (m)
Less than 65 km/h	3.0
Greater than 65 km/h	3.5
Curve with radius less than 250 m	Curve widening of 0.5 m per lane
Shuttle flow with active control	3.5

Note to Table 4-2: Where existing lane widths are narrower than those specified in the table above, the lane widths may be maintained. However the TMP must document provisions for turning paths, sight distances and vulnerable road users.

Where fixed vertical obstructions such as fences or safety barriers are restricting the available shoulder width, consideration should be given to increasing lane widths to accommodate larger vehicles.

See [Section 4.4.5 Heavy vehicles](#) for additional information regarding lane width considerations for heavy vehicles.

4.3.3 Roadside design

Roadside design aims to provide the safest transport network for all users by minimising the likelihood and severity of vehicles colliding with other road users, objects or other hazards within available funding constraints. It involves the consideration of the existing roadside environment and the future environment after the temporary traffic management measures have been implemented.

Previously, the clear zone concept was used to define a specific area beside the road to be evaluated for roadside safety and it was considered that hazards outside the clear zone were acceptable. This is no longer appropriate as the risks and hazards cannot be eliminated no matter how wide the clear zone is.

All hazards in the road reserve, both adjacent to the road shoulder (or kerb) and within median, must be considered in terms of the level of risk they pose and whether appropriate mitigation measures should be implemented.

A clear area on the roadside is one of several possible mitigation measures. Other examples include:

- Improving the delineation of the road;
- Changing the cross-section (including the use of wide medians and median barriers);
- Installing temporary safety barriers;
- Removing the hazard;
- Relocating the hazard to a position where it is less likely to be struck;
- Reducing the impact severity posed by the hazard; and
- Accepting the risk of the untreated hazard where the frequency of hitting the hazard and severity are both low. In this case, the risk should be monitored.

The most appropriate solution for a work site, which might be a combination of options listed above, must be determined based on a documented risk assessment and detailed in the TMP.

4.3.4 Minimum clearances of workers to traffic

Work must be planned and designed to provide maximum achievable clearances to workers on foot and plant.

When performing static or dynamic work, the minimum allowable clearance of 1.5 m must be maintained between workers on foot, plant and traffic, unless the following is demonstrated:

- A site or activity specific risk assessment must be completed, demonstrating that the hierarchy of controls is applied and all higher order controls have been considered;
- Site specific controls for works, including the minimum controls detailed in [Section 4.3.5 Protection of work area](#), have been determined and detailed in the traffic management plan; and
- Work specific minimum controls, detailed in [Section 7.8 Designing a TGS: Dynamic work](#) have been determined and detailed in the traffic management plan.

4.3.5 Protection of work area

In accordance with [Section 4.3.4 Minimum clearances of workers to traffic](#), all work areas should be distanced as far away from through traffic as possible via the use of *around* or *past* traffic management

methods. Where work is required to be performed near traffic, [Table 4-3](#) provides the mandatory controls and additional recommended controls for the protection of a work area.

Table 4-3. Mandatory and recommended controls for protection of a work area

Distance of work area to traffic	Mandatory and recommended controls			
	Mandatory/ recommended	Static work		Dynamic work
		Work duration greater than 4 weeks	Work duration less than 4 weeks including short-term work	*Continuous and frequently changing work
Closer than 1.5 m	Mandatory controls	Temporary safety barrier	- Delineation of work area - Speed zone of 45 km/h or less	- Speed zone of 45 km/h or less - Shadow vehicle
	Recommended controls	Speed zone of 85 km/h or less	- Speed zone of 35 km/h or less - Temporary safety barrier	- Delineation of work area - Speed zone of 35 km/h or less
Between 1.5 m and 3 m	Mandatory controls	- Temporary safety barrier where speed zone is greater than 75 km/h - Speed zone of 65 km/h or less where no temporary safety barrier is used	- Delineation of work area - Speed zone of 65 km/h or less	- Speed zone of 65 km/h or less - Shadow vehicle
	Recommended controls	- Delineation of work area - Temporary safety barrier where speed zone 85 km/hr or less	Temporary safety barrier	- Delineation of work area - Speed zone of 55 km/h or less
Between 3 m and 6 m	Mandatory controls	Speed zone of 85 km/h or less where there is no safety barrier	- Delineation of work area - Speed zone of 85 km/h or less where there is no safety barrier	Speed zone of 85 km/h or less
	Recommended controls	Temporary safety barriers	Temporary safety barrier	- Delineation of work area - Speed zone of 65 km/h or less
Greater than 6m	Mandatory controls	Worker symbolic (T1-5) sign when workers are visible to road users	Worker symbolic (T1-5) sign when workers are visible to road users	As per Section 7.8.
	Recommended controls	- Delineation of work area - Temporary safety barriers	Delineation of work area	Delineation of work site

Notes* to Table 4-3:

1. *Minimum controls for intermittent work are provided in [Section 7.8.5.1 Minimum controls for intermittent work](#).*
2. *All work sites protected by an approved temporary safety barrier must meet the requirements of [Section 6.7 Temporary safety barriers](#).*
3. *Where a temporary safety barrier is not installed, work site delineation should be provided by way of cones, bollards or similar in accordance with [Section 6.8 Traffic guidance and delineation devices](#).*
4. *Where a temporary safety barrier has been installed as a recommended control, the mandatory speed zone reduction may be re-assessed in accordance with the acceptance conditions provided on the Transport [Safety Barrier Products website](#).*
5. *If there is a risk of workers, plant or equipment encroaching into the through lane of traffic, containment fencing in accordance with [Table 6-13](#) should be installed.*
6. *In addition to the controls detailed in [Table 4-3](#), signage must be installed for the relevant works in accordance with [Section 6 Signs and devices](#).*

When determining the distance between a work area and traffic, the distance must be based on the location of the closest workers or object to the edge of a travel lane (the lane line), including bicycle lanes.

The edge of the traffic lane is the outside limit of the traffic lane, usually defined by an edge line or the kerb. If there is no edge line or kerb on the road (i.e. such as a rural road) then the edge of the traffic lane would be edge of the sealed pavement for a sealed road or the drainage swale for an unsealed road.

4.3.6 Sight distances

Suitable sight distances enable road users to perceive and react to a hazardous situation on the road ahead resulting in safer and more efficient traffic management. When designing a TMP, the PWZTMP qualified person must ensure adequate sight distance is maintained at all times along the alignment including at intersections and accesses. The following sight distance checks should be considered:

- Stopping Sight Distance (SSD);
- Safe Intersection Sight Distance (SISD);
- Overtaking Sight Distance;
- Intermediate Sight Distance;
- Approach Sight Distance (ASD); and
- Pedestrian Sight Distance.

Detailed steps and calculations associated with assigning appropriate sight distances are described in detail in Austroads *Guide to Road Design Part 3 and Part 4a* and Transport Supplement to AGRD Part 3 and Part 4a. Advice on sight distance can also be provided by Road Design and Traffic Engineering.

Sight distances must be maintained to all safety elements of the work site including but not limited to:

- Traffic control devices;
- Queued vehicles;
- Signs and delineation;
- Site access and egress points;
- Intersections;

- Private and commercial accesses;
- Merge tapers;
- Work vehicles; and
- Pedestrian, cyclist and other vulnerable road user paths and crossing points.

Best practice is to provide as much sight distance as possible, while considering drivers perception of the upcoming hazards, however it is often restricted by the following:

- Road geometry including horizontal and vertical curves in road;
- Environmental factors such as:
 - Weather;
 - Vegetation; and
 - Time of day.
- Road user factors including:
 - Traffic queues;
 - Types of road users; e.g. over-dimensional vehicles, motorcyclists; and
 - Road user behaviour such as vehicles closely following each other.
- Local site features, both temporary and permanent, including the presence of the following located in the median or on the roadside:
 - Safety barriers and kerbs;
 - Signs;
 - Street furniture, including landscaping and fencing;
 - Steep cuttings;
 - Parked vehicles;
 - Railway crossings; and
 - Bridges.

It is important to note that stopping sight distance for heavy vehicles varies from that of other vehicles. Information regarding stopping sight distance for heavy vehicles can be found in the [Performance-Based Standards Scheme – Network Classification Guidelines](#).

Additional sight distance requirements are provided in [Section 7.8.5.2 Sight distances for intermittent work](#).

4.3.7 Traffic staging plans

Traffic staging plans (TSPs) are typically road design drawings that show how traffic passes safely around, past or through the work area during various stages of a roadwork project.

A road designer must design the horizontal and vertical alignment of the temporary roadway. In addition and where detailed, a PWZTMP qualified person must design the layout for the temporary signage and delineation. These activities may be completed by two people or one person with both qualifications. Accordingly, the alignment and layout must be signed off under road designer and PWZTMP competencies.

The TSP and any supporting TGS must show the alignment, signage and delineation, and allow for easy installation and checking. As such, it may be necessary that the TSP be separated from other staging plans, such as a construction staging drawings.

A TSP must show the following details (where applicable):

- Lane configurations on existing and new (temporary and permanent) pavements;
- Intersection layouts and temporary traffic signals arrangements;
- All existing and temporary traffic signage, delineation and pavement markings;
- Footways and cycle ways;
- Bus stop locations;
- Work areas;
- Construction site ingress and egress locations including acceleration and deceleration lanes (or shoulders);
- Access to local properties and side roads;
- Temporary street lighting;
- Temporary barriers; and
- Chainages.

Other factors that should be considered (that may form part of a staging construction design) when designing the TSP include:

- Temporary pavement design;
- Temporary drainage design, including erosion and sedimentation control;
- Longitudinal and cross-section plans of temporary alignment;
- Utilities and their impact on staging; and
- Temporary retaining structures.

4.3.8 Detours

A detour must be carefully planned at the TMP stage such that:

- The alternative route is capable of accommodating all classes of vehicles that are to be detoured in terms of mass, height or width limits of any structures;
- Stakeholders, such as the local council and police, are informed and concur with the proposed detour;
- Local residents are consulted and informed in a timely manner; and
- Consent is obtained for local and regional roads from the relevant local council in accordance with the Heavy Vehicle National Laws.

The following additional aspects should be considered when planning a detour:

- Lane widths;

- Geometric design of intersections and their capability to accommodate the vehicles that are being detoured (for example, B-double sweep paths and small roundabouts);
- Existing intersection controls and priority relative to the proposed route (i.e. will Give Way or Stop sign control be in conflict with the detour route);
- Existing traffic flows and turning movements;
- Vulnerable road user movements and existing facilities;
- Land use along the detour route and the need to manage any environmental impact (i.e. noise); and
- Sensitive land uses along the detour route. For example, aged care facilities, child care centres etc.

Further information regarding TGS design considerations for detours is contained in [Section 7.7.2.3 Detours](#).

4.4 Providing for specific road users

4.4.1 General

During the development of a TMP, the needs of specific road users must be understood and managed for the extent of the works. The locations of need, travel paths and desire lines must be factored into the development of the TMP to ensure safety and access is maintained.

Specific groups to be assessed include:

- Vulnerable road users including:
 - Pedestrians;
 - Cyclists;
 - Motorcyclists; and
 - Users of scooters and other similar devices/vehicles.
- Heavy vehicles, including oversize overmass vehicles.

When it is identified that TTM arrangements may affect any of the above groups, paths and routes must be maintained in an acceptable and hazard-free condition at all times. Restriction or closure of routes, accesses or desire lines must be determined via appropriate consultation and approval.

4.4.2 Pedestrians

It is essential to understand the type, need and access required for pedestrians affected by the works. Pedestrians include high risk pedestrians which are at greater risk to hazard such as persons with a disability, the elderly and children.

Information that should be obtained includes:

- Existing pedestrian paths within and adjacent to the project boundaries, e.g. desire lines, key routes, attractors, destinations or linked trips;
- Pedestrian accident history;
- Surrounding land use, including access needs to businesses, schools, pubs or community facilities;
- Expected numbers and types of pedestrians; and
- Pedestrian accessibility requirements e.g. shopping trolleys, prams, motorised or non-motorised wheelchairs or walking frames.

The TMP must ensure pedestrian needs are addressed, including:

- Specific requirements for high risk pedestrians e.g. persons with a disability (particularly with vision or hearing impairment), children or elderly; and
- Provision of additional signage, guidance (e.g. pedestrian fencing) and protection (e.g. crash barriers and crash cushions) in high risk locations.

Additionally, the TMP must determine the need for a road safety audit.

Generally, pedestrians will take the most direct route possible. At work sites where it has been necessary to introduce temporary arrangements for pedestrians, movement patterns of pedestrians must be observed, especially during the first week, and necessary changes made to ensure that pedestrians are able to move safely. Observations should cover both day and night conditions as required by the road safety audit process, and any amendments should be made to the approved TMP.

Key considerations when planning for pedestrians are provided in [Table 4-4](#).

Table 4-4. Planning considerations for pedestrians

Consideration	Requirement
Defining the work area	The work area must be clearly defined by using an appropriate fencing. Where pedestrians have been diverted onto an existing roadway, the pedestrian path must be separated from vehicular traffic by a mesh fence at a minimum, provided that there is sufficient width for the anticipated pedestrian volumes, storage and flow directions. Where these conditions cannot be met, a safety barrier or an alternative pedestrian path must be provided or the work site redesigned to pass a road safety audit based on pedestrian access. Pedestrians must be separated from any trenches or the trenches covered with appropriate plates until backfilled.
Reversed traffic direction	Special safety requirements may be needed where traffic is required to travel temporarily in the wrong direction, at a detour, crossover or other location. In particular: • Containment fences may be required to control the crossing point; • Personnel may need to be positioned to guide and assist the pedestrians; or • LOOK BOTH WAYS, TWO-WAY TRAFFIC (T8-5) signs may be required, especially where one carriageway of a divided road is closed. The signs must be placed on both sides of the open carriageway facing pedestrians.
Footpaths	Temporary footpaths must provide a clear path of travel and must be: • Adequately signposted to indicate the direction of the footway; • Of all weather standard including ramps over gutters that can have water velocities above 1 m/s; • Of equivalent material and performance to the adjacent footpaths and not pose a trip hazard for the range of pedestrians; and • At local constrictions, not less than 1 m width. Elsewhere a width of at least 2 m must be provided and any additional width to aid stopping sight distance to all road users. Care should be given to maintain good conditions and widths at points of concentration of pedestrians, such as in front of shops, schools at bus stops, light-rail stops, storage islands, medians and refuges.

Consideration	Requirement
Pedestrian crossings	Care should be given to maintaining pedestrian crossing facilities and associated signs during the works. If access to existing crossings cannot be maintained, alternative facilities as near as possible to the established crossing and (to at least the same standard) must be provided. In some cases, on long-term works, grade separated structures, such as pedestrian bridges or subways that 'span' both the work area and the travel lanes, may be appropriate. Pedestrians must be prevented from crossing, where: • Medians, refuges or other physical devices separate lanes of traffic flowing in the same direction; • Traffic speeds are higher or the drivers will not be alerted to pedestrians; • There is insufficient safe congregation space; or • Stopping sight distance is not available to all road users. In these circumstances active control must be implemented, using one of the following: • Pedestrian-actuated traffic signals; • Use of portable traffic control device; or • Direct pedestrians through a chicane so that they face the direction of oncoming traffic with the required stopping sight distance before stepping on to the traffic lanes.
Lighting and security	Where it is necessary to temporarily divert pedestrians, the path must: • Have lighting provided for pedestrian crossings; and • Have all security risk areas identified and controls implemented to manage this risk. This may include provision of additional lighting, security cameras or change of design to eliminate high potential locations.

4.4.3 Cyclists

Transport is committed to the use of cycling as a transport mode, and to the provision of safe and convenient cycling facilities. Therefore, ongoing provisions for cyclists for the duration of the works must be identified and determined during development of the TMP. The following items should be considered when making this determination:

- Existing volumes of cyclists;
- Existing cyclist facilities;
- Available width and alignments;
- Traffic speeds and volumes;
- Duration of work;
- Surface material and condition;
- Environmental effects; and
- Costs.

Following the determination to provide for cyclists, and documented in the TMP, a smooth shoulder, with a minimum width of 1.2 m (2 m required if adjacent to traffic lane with speeds above 60 km/h) and regular sweeping of the riding surface must be provided. Consideration should be given to increasing the width of the cycle path if site based factors such as steep grades or vegetation are present.

Cyclists will take the most direct route possible. At work sites where it has been necessary to introduce temporary arrangements for cyclists, movement patterns of cyclists must be observed, especially during the first week, and necessary changes made to ensure that cyclists are able to move safely. Observations should cover both day and night conditions as required by the road safety audit process, and any amendments should be made to the approved TMP.

Table 4-5 provides considerations that should be applied when planning for works in relation to cyclists.

Table 4-5. Considerations when planning for works impacting cyclists

Type	Considerations
General principles	- On down-grades, cyclists can travel at speeds of up to 50 km/h and surfaces must be able to be ridden on safely at this speed; - Many cycles have no suspension making rough surfaces unpleasant and potentially dangerous to travel on; - Provision should be made to 'feather' the edges of temporary surfaces to remove any hazardous edges; - Roadworks signs should be placed above the head height of cyclists; and - Signs and devices should be placed so that they do not force cyclists away from safe travel paths.
Grooved roads	Milling of roads prior to re-surfacing can present a hazard for cyclists. Where road surfaces are grooved, cyclists should be warned by the display of CYCLE HAZARD GROOVED ROAD (T2-207n) signs on all approaches.
Lighting of work sites	Where works affecting cycle ways are carried out for a period exceeding one day, the works must be made sufficiently visible for night time travel. That is, cyclists must be able to observe site conditions under low ambient light, including temporary access paths, and take appropriate action. In addition, as a general principle, lighting on temporary access paths should not be less than the existing level on the original path.

4.4.4 Motorcyclists

Motorcycles (and mopeds) travelling through work sites require additional consideration, particularly in terms of the road surface provided. When designing a TMP, it should be considered that motorcycles do not handle in the same way as cars, they are less stable than other vehicles on loose and slippery surfaces and are unable to brake heavily on curves.

Signage for the site must adequately identify road surfaces that are different to that on the approach. Particular hazards to motorcyclists include unsealed surfaces, some painted road markings when wet, unexpected sections of wet roads and loose gravel, especially in areas where braking and turning is required.

Table 4-6 provides considerations that should be applied when planning for works in relation to motorcyclists.

Table 4-6. Considerations when planning for works impacting motorcyclists

Consideration	Requirement
Road surfaces	Surface changes can be critical to motorcycle stability and so, surface changes must be signposted well in advance so that a motorcyclist is not forced to take rapid or unexpected evasive action. This may include: • Use of the GRAVEL ROAD (T3-13) sign where sections of sealed roads are temporarily reduced to gravel; • Use of the SLIPPERY symbolic (T3-3) sign to indicate a wet surface; or • Use of appropriate signage, such as Temporary Portable Rumble Strip Ahead Symbolic (T5-210n) sign or Boom Barrier AHEAD (T1-272n) sign to warn of specific changes.
Grooved roads	Milling of roads prior to resurfacing can present a hazard to motorcyclists. Where road surfaces are grooved, CYCLE HAZARD GROOVED ROAD (T2-207n) signs must be erected on all approaches. Where grooving is carried out on roads with curved alignments, it may be necessary to impose roadwork speed zones to help improve motorcycle safety.
Other hazards	• Materials, installation and maintenance of steel plates used to cover excavations must comply with Transport QA Specification M209, Road Openings and Restoration. • Water which drains from roadworks, should not run across traffic lanes to a depth greater than 5 mm.

4.4.5 Heavy vehicles

Transport is committed to providing safe and efficient access for heavy vehicle operation in delivering freight.

OSOM loads may travel on the road network under Notice with a gross mass up to 115 tonne and dimensions up to 5 m wide, 5 m high and 30 m long. OSOM vehicles may have a trailer axle ground contact width up to 4.6 m. Transport has no visibility of when, where or the frequency of vehicles operating under Notice, so work sites must cater for these dimensions at a minimum if work is carried out on an identified heavy vehicle route (see [NSW OSOM Load Carrying Vehicles Network Map](#)). Vehicles that exceed any of the above mass or dimension limits are required to obtain a permit via the National Heavy Vehicle Regulator.

When developing a TMP, the considerations provided in [Table 4-7](#) for heavy vehicles such as OSOM and Restricted Access Vehicles (RAV) accessing the network under notice or permit, must be identified and detailed in the TMP.

Table 4-7. Heavy vehicle TMP provisions

Consideration	Detail
Number and type of vehicles	Number and percentage of different types of heavy vehicles, time of day access requirements.
Vehicle clearances	• Length, mass, height and width of vehicles expected to pass through the site; • Minimum ground contact height required for vehicles; and • Turning path and sight distance requirements.

Consideration	Detail
Access through site	• Identification of alternate routes around work site nominated if vehicle cannot be accommodated through the site; • Works designed so that the installation of temporary barriers, contra flow arrangements and falsework structures do not significantly affect the movement of heavy vehicles; and • Location and placement for emergency stopping/pull over bays within long work sites that will accommodate long vehicles.
For higher volumes of OSOM vehicles	• Ensure Livetraffic.com.au is updated to show the works and any relevant information relating to OSOM access; • Provide a site contact name and contact details to the Transport Road Access Management unit to assist OSOM operators in negotiating the work site; • Where high volumes of heavy vehicles are expected, additional safety measures such as traffic control and VMS boards should be provided to give advanced warning to OSOM loads of the upcoming works; and • Provision of OSOM pullover areas prior to the works.

Heavy vehicles between 3.5 m to 6 m wide are accompanied by at least one escort vehicle. Escort vehicles function as a warning to other road users of an approaching oversize load. Loads in excess of 6 m wide and/or 40 m long are accompanied by NSW Police escorts.

Escort drivers must not conduct traffic management functions (aside from leading or following their OSOM vehicle) while travelling through a work site that is under independent traffic management and control. As such escort vehicle drivers must not assist in managing other road users to facilitate access of an OSOM load through a work site. This requirement does not apply to Police when acting as an escort.

Pilot vehicles which are operating in accordance with the TMP for the particular work site are permitted to manage other road users to facilitate access of an OSOM load through the particular work site.

When developing a TMP, the project location should be identified on the [Restricted Vehicle Access Maps and Lists](#) to determine if the project will be within any approved routes and network restrictions for use by heavy vehicles. The interactive RAV maps provide details on the roads and zones approved for the following heavy vehicle combinations when selecting the corresponding option, including:

- 4.6 m high vehicle routes;
- B-doubles routes; and
- Road Train Routes.

The published maps are the legally enforceable network in NSW. Freight Branch (email freight@transport.nsw.gov.au) can provide support and advice regarding types of vehicles under Notice or permits to access the network.

Where a work site impacts on the dimensions of the existing road environment, by reducing lane width or introducing temporary safety barriers, Live Traffic must be used to notify road users, including heavy vehicle operators, of the work sites. At a minimum the information on Live Traffic should include details of any OSOM restrictions and a site contact and all information provided should be clear, concise and relevant.

For assistance of advice regarding OSOM routes, site provisions or to assist in sending communications about works to the heavy vehicle industry, the Transport OSOM Access Manager can be contacted on 1300 656 371.

4.5 Speed zones

4.5.1 General

Roadwork speed zones are used to assist in managing the risk to road workers or road users in case of a hazard on or surrounding the road. This may include, but is not limited to the following:

- Where traffic travels past or through a work site or where workers are undertaking activities in or adjacent to the roadway;
- During an accident or an emergency;
- During situations of reduced visibility;
- Where loose material or stones are on the road surface, e.g. during spray sealing works;
- Where there is a reduction in quality of the road surface condition, including where vertical or horizontal alignment is inconsistent with the adjacent length of road;
- When traffic is diverted onto an opposing carriageway, detour or side-track;
- When controlling traffic in accordance with Section 5.4 Traffic control;
- Where road users are required to travel adjacent to excavations; and
- On bridges, where a reduction in the impact loading caused by traffic is required for structural safety.

4.5.2 Selection

When determining the need for a roadwork speed zone, the PWZTMP qualified person must ensure the selected speed zone is:

- Credible;
- Placed in accordance with the requirements of this Section;
- Placed to maximise the effectiveness of the road environment, i.e. topography and geometry;
- Not used alone, but used with other signs or devices as dictated by and in response to the site-specific conditions;
- Within the minimum length needed for the protection of workers and the travelling public;
- Not used in the place of more effective means of traffic control, but to complement such controls; and
- Only used for the duration of the need, and not used while work is not being undertaken or when road conditions have resumed to their normal operation.

The following factors should be considered when selecting the roadwork speed limit:

- Number and degree of vehicular and pedestrian and other vulnerable road user conflicts;
- Type and extent of the work in progress;
- Characteristics of the road;
- Proximity of workers to passing traffic; and
- Number and type of work vehicles entering or leaving the road.

Roadwork speed zones must be selected in accordance with the conditions provided below, and can be supported by using appropriate signs and devices according to the criteria provided in Table 4-8.

Table 4-8. Selection criteria of roadwork speed zones

Speed zone	Selection Criteria
80 km/h	The speed limit must be reduced to 80 km/h where: - Workers on foot, or operating plant, are between 3 m and 6 m of a traffic lane with no intervening physical barrier; - There are changed traffic conditions on the site such as, reduction in the number or width of lanes and varying surfaces; or - A transition zone is required in 110 km/h zones where a 60 km/h or a 40 km/h roadwork speed zone is used and the use of a Speed Limit AHEAD (G9-79) sign is considered inadequate.
70 km/h	The speed limit may be reduced to 70 km/h where: - Variable Speed Limit Signage is in place; - Integrated Speed Limit and Lane Use Signs (ISLUS) are in place; or - VMS are used to display regulatory speed zone signage.
60 km/h	The speed limit must be reduced to 60 km/h where: - Workers on foot, or operating plant, are between 1.5 m and 3 m of traffic with no intervening physical barrier; - Traffic control is used; - There is frequent interaction between work vehicles and through traffic; - There is a reduced standard of alignment due to the works; or - There is a loose surface such as gravel or a newly sprayed bitumen seal.
40 km/h	The speed limit must be reduced to 40 km/h where: - Workers on foot, or operating plant, are closer than 1.5 m to traffic with no intervening physical barrier; - There is a severe change in the alignment considering the surrounding speed environment; or - A bridge deck has an inconsistent surface or there might be structural damage to the bridge by vehicles travelling at higher speeds.
30 km/h*	Traffic should be reduced to 30 km/h where: - Workers on foot, or operating plant, are closer than 1.5 m to traffic with no intervening physical barrier; - The existing posted speed limit of a road is 45 km/h or less; or - It has been identified by divisional procedures.

Note* to Table 4-8: refer to Section 4.5.3 for 30km/h speed zone conditions of use.

4.5.3 Criteria for use

Priority must be given to providing safe clearances between workers and traffic through the use of other controls, rather than installing a speed limit of less than 45 km/h.

A 40 km/h roadwork speed zone should not be left in place for extended periods of aftercare, such as over a weekend. The extended use of 40 km/h roadwork zones when the work site is clear of active workers diminishes the credibility of work sites, leading to increased rates of non-compliance.

A 40km/h speed zone may be implemented at an unattended work site for more than 24 hours, provided the one-up manager to the PWZTMP qualified person or Project Manager has reviewed and approved the plan to do so. This approval must be specifically documented in the relevant TMP and TGS. Otherwise, a 40 km/h must not remain in place when the work site is not active or unattended.

A 30 km/h temporary speed zone is an additional measure to protect workers on foot in close proximity to live traffic only, and must not be used as a justification to reduce any other TTM requirement that is stipulated in this Technical Manual. As such, all temporary traffic management must be designed to comply with a temporary speed limit of 40 km/h or above.

The following conditions must be met prior to approval and implementation of a 30 km/h speed zone:

- The length of zone must not be less than 100 m or exceed 200 m;
- The speed zone must only be implemented while workers on foot or plant are active within 1.5 m and clearly visible to road users; and
- A higher temporary speed must be reinstated during any period of inactivity, such as during lunch or other breaks.

A 30km/h speed zone must not be left in place for any period of aftercare.

It is critical for these requirements to be met, to maintain the credibility of work sites and to support the effectiveness of this additional measure for protecting workers on foot. It is recognised that the slower the speed limit, the less compliant road users often become unless an immediate risk is obvious. See [Section 4.7 Credibility of work sites](#).

4.5.4 Duration

Signs showing roadwork speed limits (such as R4-212n) must only be displayed when the speed zone authorisation (SZA) applies and for the duration of the need. The duration of need is typically:

- While active works are underway and the site is attended; or
- While warranted by changed road conditions.

Outside of restriction and need, roadwork speed limit signs must be removed or covered appropriately.

4.5.5 Implementation

When implementing a temporary speed zone, the conditions in [Table 4-9](#) must be met:

Table 4-9. Speed zone implementation conditions

Aspect	Condition
At the start of zones	Roadwork Speed Limit (R4-212n) signs are to be erected on both sides of the carriageway. Where this is not possible a second sign is to be erected 0.5D from the start of the zone.
Repeater signs	Repeater signs (R4-212n) must be erected at a maximum spacing of 500 m. They are also required where traffic enters from a side road within a roadworks speed zone.
End of zone	At a distance no greater than D from the work area, a Speed Limit (R4-1) sign must be installed showing the pre-existing speed limit prior to or adjacent to the End Roadwork (T2-16 or T2-17) sign.
Conflict with other signs	When a roadworks speed zone is introduced, speed restriction signs or markings and advisory speed signs in the zone which show conflicting speeds, must be covered or removed.

Aspect	Condition
Sign size	Roadwork Speed Limit (such as R4-212n) signs used on all roads where the existing limit is 60 km/h or more must be at least 'B' size.
Erection of signs	Roadwork speed restriction signs must be erected: - Static signs – within 5 m of the edge of the outer travel lane; and - VSLs signs – within 10 m of the edge of the outer travel lane. The position of speed restriction signs must be visible to traffic, with adequate sight distance provided.

Note to Table 4-9: Requirements for the use of portable VSLs as speed restriction signs are provided in Section 6.9.2 Variable speed limit signs (portable).

Roadwork speed zones less than 65 km/h must be located such that the zone commences no closer than 100 m before the start of the work at:

- The start of the taper area;
- The traffic diversion; or
- The traffic control position (see [Figure 4-2](#)).

Where 100 m cannot be achieved, a roadwork Speed Limit AHEAD (G9-79) sign must be installed 2D in advance of the initial roadwork speed zone to advise road users of speed reduction and allow sufficient time for speed to be reduced before the reduced speed limit.

Speed zones of less than 45 km/h must be restricted to an area or areas immediately adjacent to road workers. The roadwork speed zone must end at least 50 m past the site where people are working and must not extend greater than 500 m (see [Figure 4-3](#)).

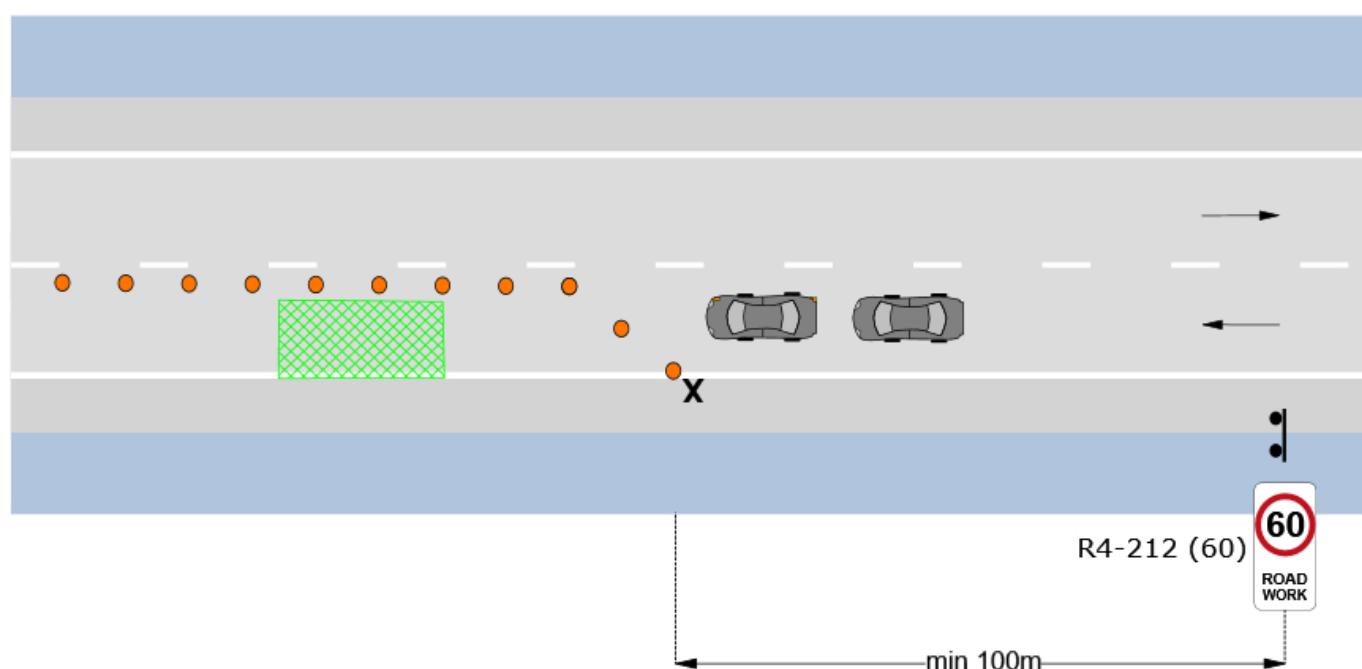


Figure 4-2. Speed zone commencement location